



**SUSTAINABLE  
TRANSPORT  
FORUM**

# **Guidelines on accessibility of electric recharging infrastructure in the EU**



**European  
Commission**

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### List of acronyms

|      |                                                                    |
|------|--------------------------------------------------------------------|
| AFIR | Alternative Fuels Infrastructure Regulation                        |
| CPO  | Recharging Point Operator                                          |
| CRPD | Convention on the Rights of Persons with Disabilities              |
| EAA  | European Accessibility Act                                         |
| eMSP | e-Mobility Service Provider                                        |
| EPBD | Energy Performance of Buildings Directive                          |
| EV   | Electric Vehicle                                                   |
| ICF  | International Classification of Functioning, Disability and Health |
| OEM  | Original Equipment Manufacturer                                    |
| STF  | Sustainable Transport Forum                                        |

## 1. Introduction

### 1.1. Target audience of these guidelines

The following guidelines are primarily aimed for **public authorities** and intend to provide recommendations and examples of good practices that these authorities can consider when setting up tenders for publicly accessible EV recharging infrastructure. While the onus of responsibility for ensuring that accessibility and inclusivity requirements is on the procurer of publicly accessible recharging points (either national, regional or local authority, car park owner, or landowner), accessibility requirements have impacts on other stakeholders involved in the provision, installation and operation of publicly accessible recharging stations:

- Original Equipment manufacturers (OEMs) – to ensure future-proof equipment that is accessible to all individuals and to promote inclusivity and equality.
- Recharging point operators (CPOs), e-Mobility Service Providers (eMSPs) and parking operators – the promotion of accessible design often leads to better overall user experiences and customer satisfaction, as well as driving innovation.
- Standardisation Development Organisations (SDOs) and in particular European Standardisation Organisations (ESOs) - play a crucial role in creating standards that ensure consistency across industries and regions. This helps in establishing a common understanding and implementation of accessibility standards in the EU. Also, adopting standardised accessibility guidelines can enhance market access for companies, making their products and services more competitive.

### 1.2. General context and objectives of the report

In the pursuit of sustainable mobility solutions, electric vehicles (EVs) have gained significant popularity in recent years. The widespread adoption of EVs relies heavily on the availability of accessible and efficient recharging infrastructure. However, for many individuals, the usability of recharging points presents a diverse range of challenges and barriers, especially for those with physical, sensory, or cognitive impairments. These barriers encompass issues related to dexterity, strength, access, and comprehension. To make the transition into electric and other alternative fuel vehicles successful, everyone should be able to take part in it. This is highly important not only from a social or ethical point of view but also from a market perspective.

According to the recitals of [Regulation \(EU\) 2023/1804 of the European Parliament and of the Council](#) (Regulation on development of alternative fuels infrastructure, commonly known as AFIR), *“Transport infrastructure should allow seamless mobility and accessibility for all users, including older persons, persons with reduced mobility and persons with disabilities. In principle, the location of all recharging and refuelling stations, as well as the recharging and refuelling stations themselves, should be designed in such a way that they are accessible to and user-friendly for as much of the public as possible, in particular older persons, persons with reduced mobility and persons with disabilities. (...) In that sense, the accessibility requirements set out in Directive (EU) 2019/882 of the European Parliament and of the Council should be applicable to recharging and refuelling infrastructure.”*

Under Article 14, AFIR requests that Member states report in the National Policy Frameworks about *“measures to ensure that publicly accessible recharging and refuelling points for alternative fuels are accessible to older persons, persons with reduced mobility and persons with disabilities in accordance with the accessibility requirements of Directive (EU) 2019/882”*. However, there is still a lot of uncertainty for different stakeholders on what will be the policies of Member States at the national level to achieve this objective.

Moreover, under Article 21, AFIR states that *“the Commission may request European standardisation organisations to draft European standards setting technical specifications for areas (...) for which no common technical specifications have been adopted by the Commission”*, which includes specifications for recharging stations to ensure access to users with disabilities as stipulated in AFIR Annex II, point 1.18 which provides the legal basis for the development of *technical specifications for recharging stations to ensure access to users with disabilities*.

These guidelines explore the crucial aspects of ensuring the accessibility and usability of recharging points, aiming to make them inclusive for all EU citizens. It is recognised that individuals may experience a combination of impairments, and as such, it is imperative that the hardware and software parameters, payment options and surrounding area of recharging points are designed with careful consideration of the diverse needs of their users, including those with disabilities.

**Within this document, both terms “accessibility” and “inclusivity” are used interchangeably. While the term “accessibility” is commonly used in AFIR and other legislation, the term “inclusivity” is used across these guidelines as an aim to ensure that people with disabilities are considered as valuable members in all aspects of society. This includes listening to their views and opinions and allowing people with disabilities to contribute to planning and provision of products and services such as EV recharging.**

Most elements and recommendations presented in this report can be applicable to both on-street and off-street parking and charging, but these guidelines refer

primarily to outdoors on-street parking. However, for some aspects of parking in covered areas the recommendations formulated in this document may be applicable in a case-by-case analysis.

Under the UN Convention on the Rights of Persons with Disabilities, to which the EU and all its Member States are party, States Parties are called to take appropriate measures to ensure to persons with disabilities access, on an equal basis with others, to the physical environment, to transportation, to information and communications (including to related technologies), , and to other facilities and services open or provided to the public, both in urban and in rural areas.

Neither EU legislation nor the UN Convention on the Rights of Persons with Disabilities set a legal definition of accessibility. For this Guidance document, the definition of accessibility provided in Zero Emission Vehicles Ireland's draft Universal Design Guidelines 2023 will be used: **'Accessibility refers to the design and provision of products, services, and environments that are usable and accessible to people with disabilities, the elderly, and those with different needs, to ensure they have equal opportunities and can participate fully in society' (1).**

Motability research (2) estimates that in the United Kingdom (3), an estimated 2,7 million people are expected to be living with disability and drive or be a passenger regularly in 2035, based on the expectation that the number of disabled drivers will grow faster (+16%) than the population (+7%) between 2020 and 2035, due to the ageing population (4). The same study expects up to half of those disabled drivers to rely on public recharging infrastructure in 2035. In Germany (5), more than ten million people are officially recognised as disabled people, of which 7,8 million are severely disabled. Similar data about the number of drivers living with disability in the EU is currently not available (assessment needed).

While we understand and recognise that all recharging stations should meet accessibility requirements that are universal, including all kinds of disabilities, ensuring this in the short term could impose complex requirements that are very challenging to meet. **These guidelines mostly encompass recommendations towards ensuring accessibility to the persons with disabilities who can drive and are allowed to hold a driving license across Member States. Yet, this is not solely a driver's concern, and future developments should expand this to consider vehicle passengers with disabilities that cannot**

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(1) Irish Electric Vehicle Charging Infrastructure – draft Universal Design Guidelines 2023

(2) Motability is an independent UK company responsible for the finance and administration of the Motability scheme, leasing wheelchair accessible vehicles, scooters and powered wheelchairs to people with disabilities <https://www.motability.co.uk>

(3) Designability's Design Guidance - Accessible EV charging 2022

(4) Electric Vehicle charging infrastructure for people living with disabilities, Final Report for Motability, 2020

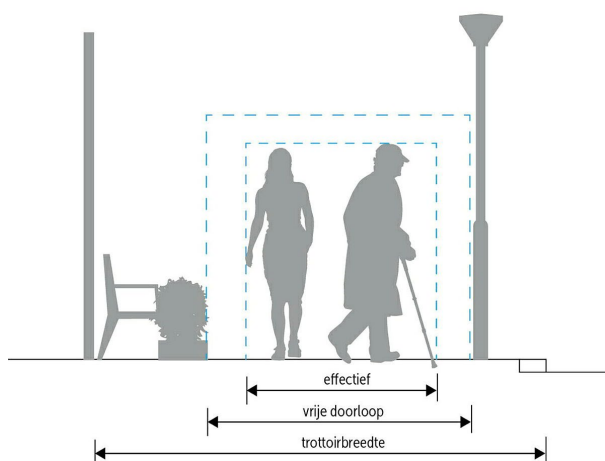
(5) NOW GmbH Guidelines for Barrier-Free Recharging Infrastructure 2023

**drive a vehicle, as well as the blind, visually impaired, or wheelchair users.** EV recharging infrastructure should also not pose an additional barrier for pedestrians using the kerbside or pavement. Some of the recommendations proposed in this document already take this into consideration.

According to the Dutch knowledge center CROW <sup>(6)</sup>, when it comes to space available for pedestrians, it is important to realize that in practice, the width of the footpath usually does not correspond to the space that is actually available for walking, as there are regularly objects that reduce the walking space available. These are usually located in a zone along the façade and/or in a zone along the pavement, but sometimes also in the middle of the footpath. The following are important for the standardisation of the dimensions:

- **The footpath width** - This is the full width from the façade (or the property boundary) to the kerbside.
- **The free passage space (clear width)** - This is the total footpath width minus the space occupied by existing objects. This space is 'free', but in practice it cannot be fully used for walking because there are 'startle distances' to walls and objects and 'elbow room'.
- **The effective walking space** - This is the part of the free passage space that can be used effectively for walking.

**Figure 1: Schematic representation of the distribution of the free and effective passage space**



Source: CROW.

Schematic representation of the distribution of the free and effective passage space

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(6) Safe and accessible pavements at charging stations and charging cables

Although accessibility features are beneficial to all citizens, the market share of vehicles for people with disabilities is small percentage of the total vehicle market but nevertheless quite significant numerically. For example, 1.2 million vehicles were in the disabled tax class in the UK at the end of 2021, representing 3% of all licensed vehicles<sup>7</sup>. Accessibility of recharging infrastructure needs the attention of various stakeholders: infrastructure manufacturers, research institutions, engineering departments, healthcare organizations, public authorities and concerned persons with disabilities themselves to create better, more sustainable, and accessible solutions. This transition must be supported and guided, including through enforcement of any mandatory requirements, by political institutions to make sure the rights and needs of persons with disabilities are considered and respected.

A crucial element to consider is that the concept of disability is related to the barriers that exist. Accessibility is about removing such barriers created by society. It is also clear that accessibility needs are extremely wide ranging. There are several types of disabilities, and it is important not to assume that one accessible solution will meet the needs of all disabled people. According to the Convention on the Rights of Persons with Disabilities (CRPD), the model of disability moves away from the medical perspective and evolves towards a human rights approach. “Persons with disabilities include those who have long-term physical, mental, intellectual or sensory impairments which in interaction with various barriers may hinder their full and effective participation in society on an equal basis with others”. The International Classification of Functioning, Disability and Health (ICF) defines the following classifications of disability types: mobility/physical, spinal cord, head injuries, vision, hearing, cognitive/learning, psychological and invisible. The largest groups impacted by current EV recharging infrastructure are:

- People with reduced mobility
- Those over pension age
- People with long term pain
- Wheelchair/walking aid users

While these are the largest groups impacted by lack of access to current EV recharging infrastructure, many other groups could potentially be adversely affected for example:

- People with learning difficulties
- People who are deaf or hard of hearing
- People who are neurodiverse

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(7) <https://www.gov.uk/government/statistics/disability-accessibility-and-blue-badge-statistics-2021-to-2022/disability-accessibility-and-blue-badge-statistics-2021-to-2022>

- People who have vision impairments<sup>(8)</sup>.

The optimal solution for new technologies to be truly accessible, each one of these areas will need to be considered throughout the design process and tested accordingly.

Without specifying all user groups related to 'persons with disabilities', it is of prime importance to determine the minimum requirements which can be applied generically and in a reasonable way. In case needed, specific requirements can be specified per user group, but this requires a more thorough and specialised work starting with a clear definition of accessibility levels. Therefore, the challenge now is to develop solutions that can accommodate the breadth of access needs, to enable as many people as possible to have a convenient and enjoyable public recharging experience.

The 'Enable Final report' realised for UK Power Networks in 2022 identified key barriers to the accessibility of on-street recharging points for disabled users (see Table 1).

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(8) Guidance on improving electric vehicle charging infrastructure, Sept 2021 Abridged report.

**Table 1 – Key barriers relating to the accessibility of on-street recharging points for disabled users**

Source: UK Power Networks

|                                                |                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Variety of users                               | Lack of consideration of users with varying mobility and visual impairments.<br>Lack of consideration of the need to account for disabled motorists, disabled passengers and pedestrians in infrastructure design.<br>Lack of consideration of the requirements of individuals with respect to the minimum proximity to charge points.                                        |
| Suitability of charge points                   | Lack of visibility of recharging points<br>Lack of consideration of space required for wheelchair users around the charge point.<br>Lack of consideration of the potential for charge points to block car door access.<br>Lack of consideration of the difficulty lining up, retrieving and returning connectors.<br>Lack of consideration of the weight and length of cables |
| Built environment and infrastructural barriers | Lack of consideration of the need for cover from weather.<br>Lack of drop kerb for wheelchair users or those with visual impairments.                                                                                                                                                                                                                                         |
| Informational barriers                         | Lack of clarity and consistency in regard to language used in payment interface.<br>Lack of clear labelling/signage.<br>Lack of apps which collate information on parking and charging availability for disabled customers.                                                                                                                                                   |

To create a more accessible urban environment and in general transport networks and promote the adoption of EVs, it is important that these challenges are addressed comprehensively, including considerations on safety – not only the EV recharging infrastructure should be accessible, it should also ensure a safe operation for particularly vulnerable road users like persons with disabilities.

The present report of the Sub-group on public authorities of the Sustainable Transport Forum aims to establish itself as a support tool for public authorities and private stakeholders on the deployment of accessible and inclusive recharging infrastructure for persons with disabilities. It identifies the key issues, problems and bottlenecks in the deployment of accessible EV recharging stations as well as good practices that can be shared with public authorities and private stakeholders responsible for the roll out of recharging infrastructure. Finally, the aim of this report is to formulate the recommendations based on existing guidelines at the level of EU Member States and beyond. These are directed at the EU institutions, national and regional authorities, and other group of stakeholders with the aim to achieve maximum interoperability between member states in this area.

### 1.3. Policy Context

As part of the [Green Deal](#), the European [Commission's Sustainable and Smart Mobility Strategy](#) (SSMS) was published in 2020 and sets the foundation for how

the EU transport system can achieve its green and digital transformation and become more resilient to future crises. It is structured around three key objectives.

- **Sustainable mobility.** An irreversible shift to zero-emission mobility.
- **Smart mobility.** Achieving seamless, safe and efficient connectivity.
- **Resilient mobility.** A more resilient single European transport area for inclusive connectivity.

The outcome should be a 90% cut in transport greenhouse gas emissions by 2050, in line with the EU's requirement to achieve climate neutrality by 2050. The sustainable and smart mobility strategy therefore makes it a key priority to boost the uptake of zero and low-emission vehicles, renewable and low-carbon fuels, and related infrastructure for all modes of transport, without further delay.

Boosting the uptake of renewable and low-carbon fuels must go together with the creation of a comprehensive network of publicly accessible recharging and refuelling infrastructure, to fully enable the widespread uptake of low- and zero-emission vehicles in all transport modes. The Strategy enhances the ambition set out by [the 'Recharge and refuel' European flagship initiative](#) under the Recovery and Resilience Facility that, by 2025, of the up to 3 million publicly accessible recharging points needed by 2030 some 1 million should be installed along EU roads and related infrastructure by 2025.

As part of the ['Fit for 55'](#) package, the recently adopted [regulation on the deployment of alternative fuels infrastructure](#) (AFIR) is a key policy initiative to achieve the ambition set out in the Sustainable and Smart Mobility Strategy. In addition, the Commission's Communication was published alongside the legislative proposal in 2021 and outlined a set of supplementary measures to support the rapid roll-out of alternative fuel infrastructure.

Alongside legislative work on electromobility, the European Commission's Sustainable Transport Forum ([STF](#)) was set up in 2015 to assist the Commission in implementing the EU's activities and programmes aimed at fostering the deployment of alternative fuel infrastructure to contribute to the EU's energy and climate goals. The STF serves as a platform for structural dialogue, exchange of technical knowledge, cooperation and coordination between the Member States and relevant public and private stakeholders. STF subgroups are important facilitators when delivering policy recommendations to public authorities.

According to the SSMS the electromobility evolution should leave nobody behind: it is crucial that mobility is available and affordable for all, that rural and remote regions are better connected, **accessible for persons with reduced mobility and persons with disabilities**, and that the sector offers good social conditions, reskilling opportunities, and provides attractive jobs. The European Pillar of Social Rights is the European compass to make sure that the green and digital

transitions are socially fair and just. In this context, the topic of the present report subscribes fully in the angle of the Strategy focused on inclusiveness.

In its **Strategy for the Rights of People with Disabilities 2021-2030**, the European Commission calls on the Member States to “take into account the specific needs of people with disabilities in all policy areas”. This concerns, among other things, accessibility: “The accessibility of built and virtual environments, information and communication technology technologies (ICT), goods and services, including transport and infrastructure, enable people to exercise their rights and are a prerequisite for people with disabilities on an equal basis with others”.

A dedicated STF subgroup was established in early 2021 to function as a platform for exchange between public authorities on all matters to promote and facilitate the development of high-quality recharging infrastructure. This includes, among other tasks, discussion on how to ensure that the publicly accessible recharging infrastructure allows for “(...) seamless mobility and accessibility for all users, including older persons, persons with reduced mobility and persons with disabilities”, as stated under the AFIR recital 38. This Task Force is chaired by the Commission (the Directorate-General for Mobility and Transport) and by POLIS - the network of European cities and regions for transport innovation. The work developed so far will be highlighted in the following sections.

## 1.4. Policy Framework on accessibility

Research work was carried out to identify the most relevant policy and legislative frameworks currently in place at the EU level and beyond, that contribute to more accessible design requirements of electric vehicle recharging infrastructure and as a result, can inform this report as a minimum basis for universal design standards for EV recharging infrastructure in the EU.

When procuring goods, services and works technical specifications must be taken into account regarding accessibility criteria for persons with disabilities.

Article 42(1) of Directive 2014/24/EU<sup>(9)</sup> and Article 60(1) of Directive 2014/25/EU<sup>(10)</sup> state that: for all procurement which is intended for use by natural persons, whether general public or staff of the contracting authority, the technical specifications shall, except in duly justified cases, be drawn up so as to take into account accessibility criteria for persons with disabilities or design for all users. This means it is a legal obligation for public buyers (or utility sector entities) to take accessibility criteria into account in technical specifications. Furthermore, Directives 2014/24/EU and 2014/25/EU require that, where mandatory

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(9) <https://eur-lex.europa.eu/eli/dir/2014/24/oj/eng>

(10) <https://eur-lex.europa.eu/eli/dir/2014/25/oj/eng>

accessibility requirements are adopted by a legal act of the Union, technical specifications are, as far as accessibility for persons with disabilities or design for all users are concerned, to be established by reference thereto. Directive (EU) 2019/88228 on the accessibility requirements for products and services <sup>(11)</sup> — known as the **European Accessibility Act** (EAA) sets common accessibility requirements to ensure that persons with disabilities and older people can access products and services on an equal basis with others. It also reflects the commitment under the UN Convention on Rights of Persons with Disabilities (UNCRPD) that State Parties act appropriately to ensure accessible products, services and infrastructures. EAA does not provide specific provision regarding electric vehicle recharging infrastructure, nevertheless it contains general accessibility requirements, applicable for products placed on the market after 28 June 2025, among others for self-service terminals, user interfaces information and built environment which could be used for recharging infrastructure.

In procurement documents, one may therefore refer to these requirements set in EAA. Public buyers may in any case decide to apply accessibility requirements that go beyond the accessibility requirements set out in EU Law.

The EAA also includes voluntary accessibility requirements on the built environment used by clients of services covered by EAA Directive itself. Member States may decide to require compliance with these requirements. For further information and example for addressing accessibility in public procurement, please see Commission Notice “Buying Social — a guide to taking account of social considerations in public procurement (2nd edition)” <sup>(12)</sup>.

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(11) <https://eur-lex.europa.eu/eli/dir/2019/882/oj>

(12) C(2021) 3573 final, DocsRoom — European Commission (europa.eu)

### European Accessibility Act in details

*European Accessibility Act, directive 2019/882 contains:*

*Directive (EU) 2019/882 on the accessibility requirements for products and services — known as the European Accessibility Act (EAA) — sets common accessibility requirements and obligations to harmonise accessibility of certain product and services. As a result, it will ensure that persons with disabilities and older adults can access products and services. It also reflects the obligation under the UN Convention on Rights of Persons with Disabilities (UNCRPD) to facilitate compliance with. This legislative proposal requires adoption by the EU co-legislators and transposition at national level.*

*The mandatory accessibility requirements ensure that products and services are designed and produced in such a way as to maximise their foreseeable use by persons with disabilities and are accompanied where possible in or on the product by accessible information on their functioning and accessibility features. In addition, Annex III of the EAA sets out accessibility requirements for the built environment, which may be applied by Member States. As the obligation to include accessibility requirements in technical specifications also applies to works contracts, the Annex can provide a useful starting point for design and construction contracts.*

*Similarly, public authorities can refer in the tenders to Harmonised standards or technical specifications for product or service on accessibility requirements (developed by European standardisation organisations), including those developed under the EAA.*

*The Directive covers the following products and services:*

- Computers and operating systems
- Payment terminals and ATMs
- Ticketing and check-in machines
- Interactive self-service terminals providing information
- Smartphones

The provisions in AFIR mentioned already in the chapter 1.2, and in particular the possibility to adopt “technical specifications for recharging stations to ensure access to users with disabilities” may be fundamental to ensure that the future deployment of recharging infrastructure is based on the same characteristics and functionalities that can fulfil the needs of people with disabilities.

This may for example entail providing sufficient space around the parking place, ensuring that the recharging station is not installed on a kerbed surface, ensuring that the buttons or screen of the recharging station are at an appropriate height and the weight of the recharging and refuelling cables is such that persons with limited strength can handle them with ease.

However, many recharging points are to be deployed in urban areas where kerbs are the norm, or as part of existing parking lots, governed by local parking policy. Given these constraining factors, it is important for public authorities to both be clear about minimum requirements bidding parties will need to meet but also allow flexibility in their application and permit alternative solutions to achieving them, such as via digital solutions for controlling charger operations.

The [EN 17210:2021‘Accessibility and usability of the built environment - Functional requirements’](#) standard aims to describe basic, common minimum functional requirements and recommendations, applicable across the full spectrum of the built environment, for an accessible and usable built environment,

following the Design for All/Universal Design principles which will facilitate equitable and safe use for a wide range of users, including persons with disabilities. The functional accessibility and usability requirements and recommendations described in this standard are relevant to the design, construction, refurbishment, adaptation, and maintenance of built environments including outdoor pedestrian and urban areas.

**EN 301 549 ‘European standard for digital accessibility’** regarding user interface and others, specifies requirements for information and communications technology to be accessible for people with disabilities.

**EN 17161:2019 ‘Design for All – Accessibility following a Design for All approach in products, goods and services’**. The *Design for All* approach means design for human diversity, inclusion and equality. Its aim is to enable maximum use of products, goods and services. This CEN-CENELEC standard is a consistent way for organisations to manage accessibility through a *Design for All* approach, so that products and services can be accessed, understood, and used by all people.

## 1.5. Methodology

The elaboration of this report is based on different actions by the members of the Task Force including an EU survey, review of existing reports and guidelines in EU Member States and beyond, presentations by authorities and experts and exchanges in the Task Force, and benchmark analysis.

POLIS prepared the questions of the survey. They were gathered in an online form and distributed to a wide range of stakeholders, to which 31 respondents replied. POLIS analysed the results and drafted this report, with support and review from the European Commission’s Directorate-General for Mobility and Transport (DG-MOVE).

## 1.6. Scope of this report

When defining the scope of this report, it is important to clearly establish what are the key aspects that should be taken into consideration when it is referred to accessible EV recharging infrastructure (meaning that it is designed in a way that is inclusive and adapted for all users, including persons with disabilities). The Task Force has selected the following the most important ones on which the guidelines focus:

1. **Hardware & pole/recharging point** e.g., ensuring that the buttons or screen of the recharging point are at an appropriate height and the weight of the recharging cable is such that persons with limited

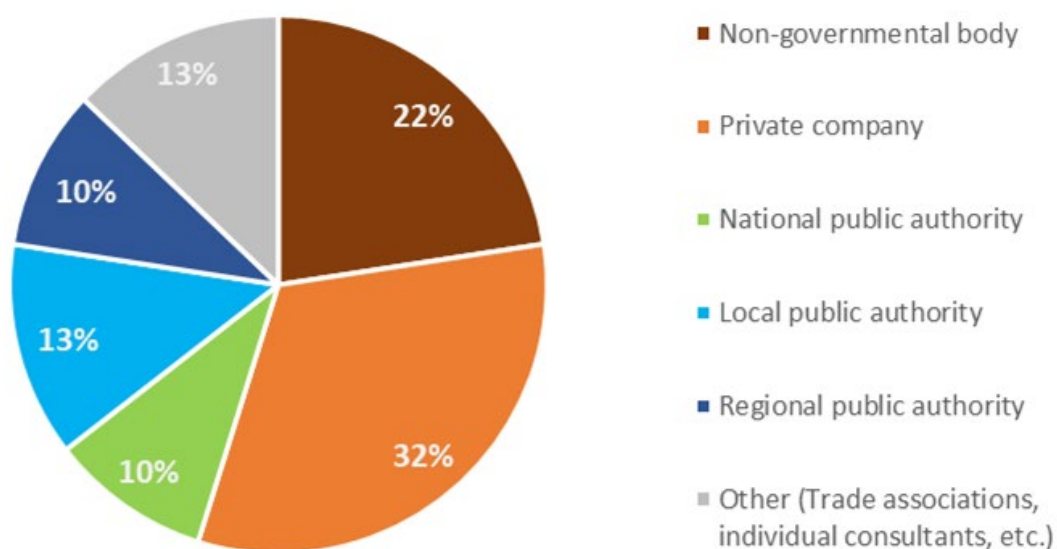
strength can handle them with ease, or a cable management system that ensures easy retractability.

2. **Associated parking spaces and surrounding environment** e.g., providing sufficient space for the parking place and around them to guarantee a safe access to and from the recharging point, or ensuring that the recharging point is not installed on a kerbed surface.
3. **User interface** e.g. ensuring that the interface of a recharging point is easy to use for people with disabilities and ensuring that consumers are adequately informed about the accessibility of a site according to AFI requirements.
4. Lastly, but not least important, the **distribution/location of accessible recharging points & associated parking spaces** as this is a fundamental issue. If not all locations of the EV recharging network are fully accessible and inclusive, appropriate metrics should be defined to quantify what should be the minimum requirements in terms of the number and spatial distribution of publicly accessible EV recharging stations which are designed according to the needs of persons with disabilities.

## 2. Analysis of the survey respondents

In the response to the survey prepared by the Task Force, 31 responses were received with around 56% from public organisations (non-governmental body, national, regional and local public authorities) and 54 % from private organisations (companies or trade organisations, consultants, etc.), as shows the following chart.

**Figure 2 – Analysis of respondents by type of organisation**



Source: Task Force 5 survey.

Analysis of respondents by type of organisation

Survey respondents from public authorities work in a variety of countries, but mostly in Western Europe including Germany (5), the Netherlands (4), Belgium (4), Spain (3), Ireland (1), Italy (1), Finland (1) and Austria (1). Figure 3 shows a graphical representation of this distribution (darker = more representation). Not all countries are represented given that there were no responses received from entities based/operating in those countries.

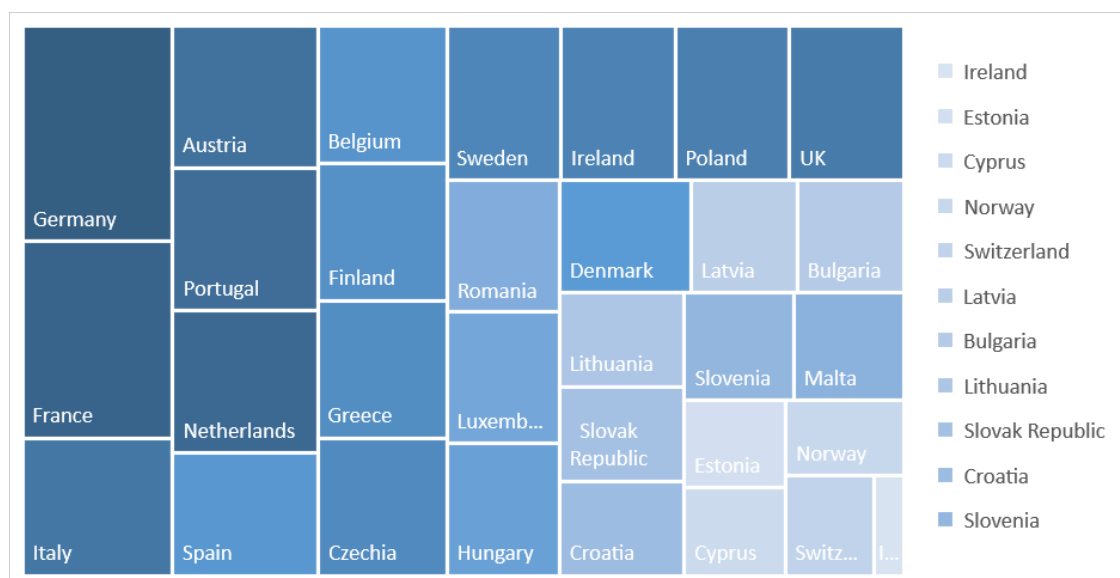
Those working in private organisations work for charge point operators (5), vehicle or equipment manufacturers (4), interest groups (3), R&D and academia (2), DSOs and TSOs (1), and other organisations including consultancies (4).

### What kind of organisations do the respondents work in?

- Operators of recharging points and networks (CPOs), end-user platforms (electromobility service providers), equipment providers (OEMs) and technology providers (TP)
- End-user and industry associations that represent and advocate the needs and requirements of their members, including persons with disabilities and EV users.
- Municipalities, regions and national governments in charge of deploying electric recharging infrastructure, master plans and/or providing funding, legal, political and strategic support.
- Academic and research organisations that work on the development, demonstration and evaluation of technologies, user behaviour, best practices, etc. linked to the electromobility sector.
- Non-profit organisations that defend the interest of people with disabilities and/or provide guidance to stakeholders along the electromobility and recharging value chain.
- And a variety of other organisations including standardisation bodies and consultants.

**Figure 3 – European countries where survey respondents are active**

Some respondents declared that their organisations worked in non-European countries, these were not included in this chart



Source: Task Force 5 survey.

European countries where survey respondents are active

Most survey respondents argued that when planning for the deployment of accessible and inclusive regarding infrastructure, the main target group should be people with physical impairments.

### 3. State-of-the-art related to accessibility of recharging points

There are currently a few key issues with recharging linked to inclusivity, these include but are not limited to:

- Poor visibility of displays and card readers.
- Kerbside not adapted for inclusive recharging, especially if there are no ramps to go on the kerbside to access the interface.
- Parking spaces too narrow, which prevent access to recharging point with a wheelchair. Areas with narrow space should have adapted design of charge points.
- Recharging stations which are not designed for a person with reduced mobility to easily lift and plug the cable to the vehicle, especially when charge port is at front of vehicle.
- Additionally, recharging stations with cables not attached, as cable handling and plugging can be a barrier.
- Lack of clear guidelines and regulations on inclusive recharging. Reusing existing regulations is key, e.g. linked to parking spaces. Local guidelines with minimal dimensions and requirements would be essential too.

#### 3.1. Challenges, bottlenecks, and limitations in regard to the deployment of accessible recharging infrastructure

The analysis of the survey's results allowed to establish the following ranking showing the concerns of stakeholders. To produce the data from the survey, a different weight was given to what the respondents put as the first three answers. Their first priority was given 3 points, their second 2 points and the third one 1 point. The score is the multiplication of all the priorities by their weights. The bigger the number, the more important the problem.

**Figure 4 – Ranking of the major problems, bottlenecks and limitations when deploying accessible EV recharging infrastructure**

| <b>Problems, bottlenecks, and limitations</b>                                                                                                                                                          | <b>SCORE</b> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Lack of common standards for accessible EV recharging (for example for recharging station equipment or parking spaces);                                                                                | <b>53</b>    |
| Absence of a clear regulatory framework at European, national, regional and local level;                                                                                                               | <b>33</b>    |
| Lack of available public space (mostly for on-street charging);                                                                                                                                        | <b>25</b>    |
| Lack of knowledge about technical developments/requirements, upcoming technologies and possible redundancy of current systems;                                                                         | <b>17</b>    |
| Lack of know-how from public authorities and charge points operators on the specific needs of people with disabilities or the elderly;                                                                 | <b>13</b>    |
| Technical feasibility of retrofitting, even if the costs would not be very high;                                                                                                                       | <b>11</b>    |
| Retrofitting costs becoming too cumbersome and not cost-efficient ("reasonable accommodation" concept according to the United Nations Convention on the Rights of Persons with Disabilities (UNCPRD)); | <b>9</b>     |
| Uncertainty about the legal 'interface' between national disability rights legislation and any future European regulatory framework;                                                                   | <b>9</b>     |
| Concern over whether the public would prioritise the funding of accessible recharging over other competing budgetary issues;                                                                           | <b>8</b>     |
| Lack of knowledge about installation and maintenance costs and longevity of equipment                                                                                                                  | <b>2</b>     |

Source: Task Force 5 survey.

Ranking of the major problems, bottlenecks and limitations when deploying accessible EV recharging infrastructure

**For clarity purposes, it is important to highlight that that these opinions and remarks do not reflect the view of the STF, being solely a collection of opinions gathered, some of which might be contradictory.**

### The survey respondents highlighted some examples of the issues that people with disabilities face when using EV recharging infrastructure

A few examples have been submitted by a **German** respondent pointing out: recharging points with a very small and not accessible display, no cables attached and poor user interaction, card readers and displays not sufficiently visible and inaccessible, parking spaces which are too narrow, or recharging points that are difficult or impossible to reach. One specific example in terms of bad practice presented a female driver unable to reach the public recharging point with a wheelchair. For this wheelchair driver to reach the recharging station, it would have been necessary to squeeze in between the connected car and the pole. The adjacent grass strip which provides access to the control panel of a recharging point was also not adapted for wheelchair users. Finally, another example of non-accessible recharging infrastructure was characterized by kerbside design preventing the access to recharging points.

One respondent also highlighted that most traditional recharging points designs and layouts do not offer enough room for those needing a wheelchair. It is also difficult for someone with mobility restrictions to lift the cable and then move to the charge port. Some vehicles have a charge port on the front, which could be particularly challenging when the charger is located adjacent to the car. On-street recharging often features a charger located up on a kerb/pavement which would require the driver to climb onto the kerb/pavement and then back down to the parking lot level, something which would be quite difficult for those with mobility restrictions.

In the **Netherlands**, one respondent identified some recharging points built on pavements with the socket facing 2 EVs and a separate pole (sign) in front of the sockets, which cannot open. Besides this if the sockets face the vehicles, access is more difficult. Additionally, another example of the dilemma between ensuring sufficient recharging infrastructure while simultaneously ensuring accessibility is the proliferation of cables potentially obstructing the pavement.

## 3.2. Specific Challenges for the design of the recharging points (hardware and interface)

Making the recharging infrastructure available for disabled persons requires adapting the hardware and interface. It means adjusting recharging station component height, plug handle, cable weight, rigidity and length, socket and connector, socket cover, screen tilt and user interface, payment methods or charging station footwell so that these are accessible.

The specific challenge here is that in the absence of mandatorily applicable standards, the parameters applied by every producer of the hardware and consequently recharging station can be much diversified from country to country resulting in the impossibility of reaching economies of scale. Therefore, a

common approach throughout the European Union is desired by both the industry and public authorities.

The requirements regarding interface can potentially overlap with those concerning interface for other uses and the elements of these are already considered in the European legislation (European Accessibility Act) and existing standards. It is here particularly important to have the possibility of hearing instructions (plugging headsets for audio outputs), even more when a recharging station is self-service.

Still, the question of height for the charging station and the interface column is specific and even if the range of height is established, it is important that the recharging station is located directly on the ground to minimise additional height differences.

The weight of the recharging cable may also increase significantly if the charging station has a high-power output (especially for fast charging). In such cases, it is important to include a cable management system if applicable.

Wireless recharging would facilitate many requirements regarding hardware and software however it is not expected to become a widespread solution in the near term.

### 3.3. Specific challenges for parking place requirements and surrounding area

According to a survey launched by Research Institute for Disabled Consumers <sup>(13)</sup> 66% of respondents felt that the hazards and barriers around the car and EV recharging points would be difficult to navigate. The organisation of a parking space and of a surrounding area in the vicinity of a recharging point with the needs of disabled people in mind can also have different background conditions if it concerns:

- (a) a recharging point at a motorway service station, on major national roads, local roads or in an urban area
- (b) A recharging point in a covered parking area under, within or above buildings.

It requires a well thought through planning of space which is not always evident if the space where recharging points are placed is not 'new' and needs to be accommodated to the existing conditions, for example a supermarket parking or

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(13) <https://www.ridc.org.uk/transport/inaccessible-charging-barrier-electric-disabled-and-older-drivers>

an existing station along a highway with a limited space for expansion. A recharging point in an urban area can also be constrained by the narrowness of a street where a recharging point is placed.

Some recharging point operators are in a position where they own both the recharging point, the immediate environment, and the surrounding areas. This gives them flexibility to find an appropriate solution for that recharging point. However, many charge points are retrofitted into areas which are not owned by the recharging point operator, limiting what recharging point operators can change in the surrounding environment. This was raised by a recharging point operator who has partnerships with supermarkets. They were constrained to standard bays due to the lack of influence in the parking policy for the site and landowners not wanting to lose any further bays. A similar issue is occurring with local authorities and on-street recharging point providers. Recharging point operators also raised the point as to who should have responsibility for certain elements of the installation, such as dropped kerbs, which may not have been required previously in an on-street location<sup>(14)</sup>.

### 3.4. Challenges linked with the spatial distribution of accessible recharging points

The question here is what principle should be guiding the deployment of recharging points accessible for disabled persons in terms of coverage and spatial distribution. While the optimal aim would be to make all recharging points accessible, in practice it can be impossible to realise in some locations, for example because of the lack of space (a narrow street; pre-existing plan of a location that cannot be easily changed) or because of the associated costs.

The situation can be however much improved and adjusted in favour of disabled persons if there is a clear public policy and specific requirements. These should require coverage of less populated, low income or less profitable areas (which is also important for social acceptability and general inclusiveness) with accessible recharging points. Here some ratios and indicators could be helpful in guiding public authorities and include the percentage of recharging points required to be accessible in a given station consisting of several recharging points.

#### Some examples in the EU and beyond

In 2017, the state of **California** published California Building Code (CBC) regulations for commercial car parks, which specifies if a facility provides electric vehicle recharging stations, then a certain number of them also need to be made

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(14) 'Electric vehicle adoption for disabled consumers', Energy Saving Trust

accessible to individuals with disabilities. In most cases, this will mean at least one recharging bay needs to be accessible and adhere to technical requirements on the size of the bay, design of the charger, required signage, and also access requirements on the surrounding environment, whereby 'paths of travel' to other public services from the charge point (such as restrooms) also should be accessible <sup>(15)</sup>.

In **Canada** the BC Building Code requires where there are EV recharging stations included in the parking area, for every 25 parking stalls with EVSE, one should be accessible. For every 10 accessible stalls, one should be accessible <sup>(16)</sup>.

**United Kingdom** Government guidelines make recommendations for 6% of off-street parking to be allocated to disabled people unless otherwise covered by local planning regulations <sup>(17)</sup>.

In 2016, the EV Association of **Scotland** released their Electric Vehicle Charging Infrastructure design guide and highlighted an important issue of EV chargers and disabled bays. A specific recommendation was made not to turn existing disabled parking spaces into shared EV spaces. Combining both of them deprives the disabled users of spacious parking spaces that are near the prime locations. This was in line with guideline examples from the **USA**, which stated that accessible EV chargers should always be in addition to existing disabled parking spaces <sup>(18)</sup>. Given that only 19% of disabled people in the **UK** hold Blue Badges, it is recommended that up to 24% of the total capacity (in addition to Blue Badge spaces) is given to accessible EV bays. The provision of Blue Badge spaces with EV charge points would only accommodate a relatively small percentage of the population with a disability, and while ICE cars are still dominant it could reduce available parking for Blue Badge holders.

When asked about what good indicators/principles/criteria should be used to ensure an **adequate spatial coverage** of accessible and inclusive recharging infrastructure, most survey respondents indicated **a preference for metrics that avoid the “one size fits all” solutions** (i.e. percentage based or area based). Most prefer either a mixed approach of these two metrics, or solutions based on specific needs and requirements from users, as well as location-based indicators defined by the competent authorities.

In regard to TEN-T corridors, a key question is “would having EV recharging places reserved for people with disabilities on TEN-T corridors be an adequate and cost-effective solution to ensure accessibility?”. To this question, the survey

(15) <https://up.codes/viewer/california/ca-building-code-2016/chapter/11A/housing-accessibility#1108A>

(16) [https://www2.gov.bc.ca/assets/gov/farming-natural-resources-and-industry/construction-industry/building-codes-and-standards/guides/2020\\_building\\_accessibility\\_handbook.pdf](https://www2.gov.bc.ca/assets/gov/farming-natural-resources-and-industry/construction-industry/building-codes-and-standards/guides/2020_building_accessibility_handbook.pdf)

(17) [https://www.britishparking.co.uk/write/Documents/Library%202016/Bay\\_Sizes\\_-\\_Jul\\_2016.pdf](https://www.britishparking.co.uk/write/Documents/Library%202016/Bay_Sizes_-_Jul_2016.pdf)

(18) <https://www.access-board.gov/tad/ev/>

respondents answered mostly no. Solutions that were offered include a booking system for accessible recharging points, finding a way to make all recharging infrastructure accessible, creating recharging points that are prioritised for people with disability, but are not reserved. In addition, they highlighted the need to have a business case for the CPOs. It also needs to be pointed out that according to AFIR, operators cannot limit the use of publicly accessible recharging points to a specific group of users. This puts severe legal constraints on the possibility of reserving certain recharging points for persons with disabilities.

### 3.5. Level of responsibility between public and private entities

Survey respondents were asked to rank from 1 to 5 the level of responsibility which should be assigned to the EU institutions, local, regional and national authorities, private operators and public-private partnerships for various segments linked to the accessibility/inclusivity of EV recharging infrastructure stated in the column on the left below.

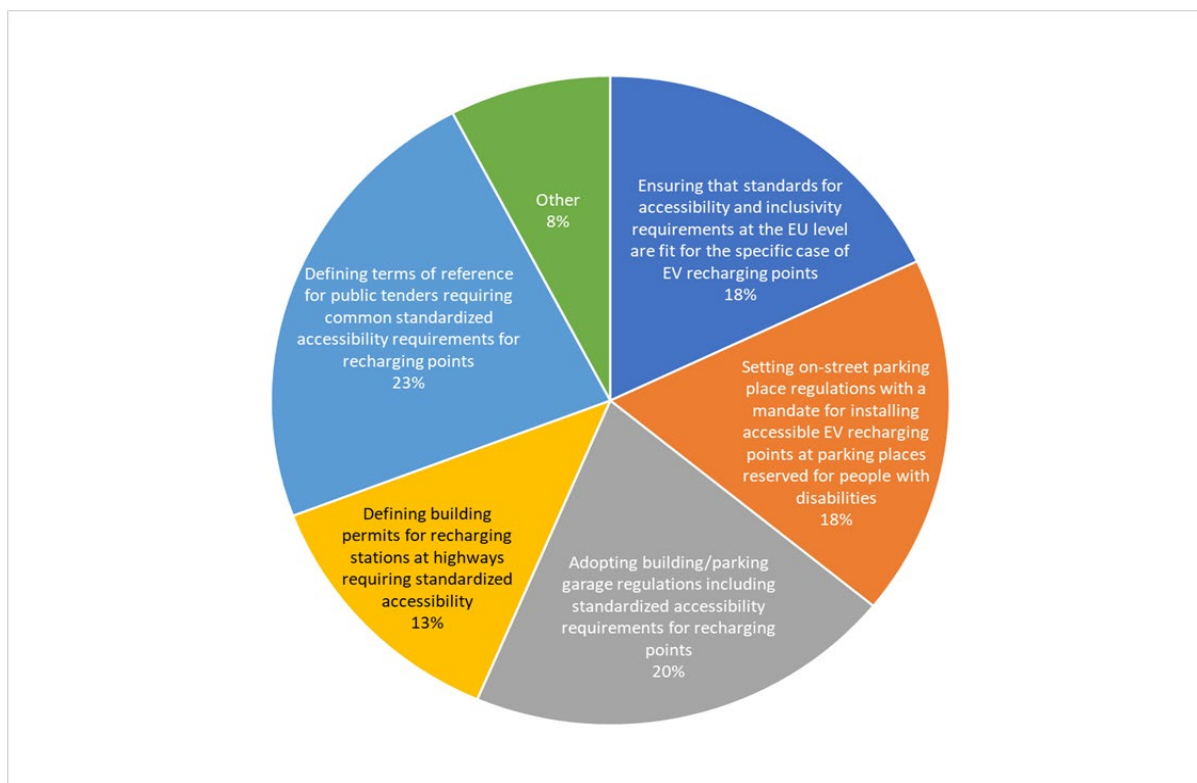
**Table 2 – Level of responsibility to be assigned to each stakeholder on the different aspects of accessibility requirements**

|                               | EU institutions | National authorities | Local and Regional authorities | Private operators | Public-private partnerships |
|-------------------------------|-----------------|----------------------|--------------------------------|-------------------|-----------------------------|
| Policy framework              | <b>4,3</b>      | 3,7                  | 2,1                            | 2,0               | 2,5                         |
| Legislation and regulations   | <b>4,3</b>      | <b>4,0</b>           | 2,5                            | 2,0               | 2,5                         |
| Physical implementation       | 2,2             | 2,8                  | 3,4                            | <b>4,5</b>        | <b>4,0</b>                  |
| Monitoring and evaluation     | 3,2             | 3,8                  | <b>3,6</b>                     | 3,3               | 3,0                         |
| Capacity building             | 2,7             | 3,4                  | 3,5                            | 3,8               | 3,4                         |
| Communication & dissemination | 3,4             | <b>3,9</b>           | 3,5                            | 3,0               | 3,0                         |

The respondents expect the most from the EU institutions and national authorities as far as policy and legislative framework are concerned. Private operators and public-private partnerships should be in charge of physical implementation and regional and local authorities ought to play the most important role in monitoring and evaluation. The communication and dissemination are to be the domain of all public authorities at all levels.

In terms of the options and tools/instruments that public authorities have, to facilitate and ensure the deployment of accessible and inclusive recharging infrastructure, the survey responses are summarised below.

**Figure 5 – Identification of different options for public authorities to facilitate the deployment of accessible and inclusive recharging infrastructure**



Source: Task Force 5 survey

Identification of different options for public authorities to facilitate the deployment of accessible and inclusive recharging infrastructure

The respondents have made several suggestions of applicable solutions or aspects to be considered (here below) to which we will refer in the chapters containing the review of best practices and recommendations:

|                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recharging infrastructure (digital and communication aspects) | <ul style="list-style-type: none"> <li>• Possibility of a touchless interaction via an application. Websites and applications involved should be accessible according to the European Accessibility Act and the relevant chapter 9 of the EN 301 549</li> <li>• The readability of the information on user interface (brightness of the screen display, contrast, size of the font, big screen buttons, easy to use for colour blind persons) and the possibility of hearing instructions (plugging headsets for audio outputs). It is particularly important if a recharging station is self-service.</li> <li>• Use of plain language and avoiding jargon (according to the European Accessibility Act the complexity of the text should not be superior to level B1 of the Common European Framework of Reference for Languages)</li> </ul> |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Urban design                         | <ul style="list-style-type: none"> <li>• Dimensions of the parking spaces adapted for disabled persons.</li> <li>• The physical environment (space to manoeuvre, lower kerb side, placement of collision barriers, etc.) are the most critical components. space requirements must in many cases be specified early in the process and can be difficult to change during the course of the work.</li> <li>• Requirements to specific site layouts should be avoided, but only be standardised as a recommended baseline to use (an operator and/or landowner local should choose the best option for a given recharging point)</li> <li>• EN 17210 standard on the built environment provides good basis such as the appropriate height and position of the recharging equipment (so it can be used by both sitting and standing persons), the access to pedestrian route and proximity to buildings and services and providing enough space around the parking space to enable manoeuvring needed to enter and leave a vehicle</li> </ul> |
| Recharging infrastructure (hardware) | <ul style="list-style-type: none"> <li>• Strict requirements should concern mainly hardware part of recharging infrastructure.</li> <li>• Adequate length and weight of a cable to facilitate its handling.</li> <li>• Weight compensation system to facilitate its handling or the use of an extendable arm that holds the connector while the driver manoeuvres to a recharging point.</li> <li>• When providing Radio Frequency Identification (RFID) readers and contactless payment functionalities, screens and connectors should be accessible at a seated height</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Other comments                       | <ul style="list-style-type: none"> <li>• It is not enough to have one accessible recharging point per x number of points as they have to be distributed evenly in public places,</li> <li>• In order to ensure coherence on the EU level and to make sure that all persons with disabilities have the same opportunities and access there should be an EU level legislation governing it.</li> <li>• A first step to advance towards an accessible EV recharging infrastructure would be to use the relevant functional accessibility requirements in Annex I and III of the European Accessibility Act as well as EN 17161:2019 'Design for All – Accessibility following a Design for All approach in products, goods and services</li> <li>• The considerations should be applied to all electric vehicle recharging points, not just those in public places.</li> </ul>                                                                                                                                                                |

## 4. Overview of existing guidelines in place across the EU and beyond

This chapter provides an overview of the main legislation and guidelines on the accessibility of recharging points in the EU and beyond, including EU-funded projects such as eCharge4Drivers and USER-CHI, as well as feedback from survey respondents on their impact.

- **Germany:** DIN SPEC 91504 for barrier free recharging infrastructure for electric vehicles).
- **Ireland:** Zero Emission Vehicles Ireland (ZEVI) electric vehicle charging infrastructure universal design guidelines.
- **Spain:** State Reference Centre for Personal Autonomy and Technical Aids (CEAPAT) accessibility criteria report for recharging infrastructure
- **UK:** Standard PAS 1899:2022, Designability/Motability Guidelines
- **Sweden:** *Stolpe in i Stad och Land* (SISL) Mellersta Norrland project
- **USA:** US Access Board Design Recommendations for Accessible Electric Vehicle Charging Stations

In **Spain** there is the CEAPAT guidelines “*Criterios de accesibilidad de los puntos de recarga de vehículo eléctrico o electrolineras*” defining a minimum of light at a recharging station and the criteria of accessibility <sup>(19)</sup>.

In **Germany** the Master Plan Charging Infrastructure II <sup>(20)</sup> was published in October 2022 as the German Federal Government’s roadmap for the expansion of charging infrastructure. Measure 39 “Barrier-free access to the charging infrastructure” provides for the German Federal Ministry for Digital Affairs and Transport (BMDV) the basis to develop technical requirements with the involvement of relevant stakeholders to enable people with disabilities to use recharging infrastructure. On behalf of the Federal Ministry for Economic Affairs and Climate Protection under the umbrella of NOW GmbH, on November 2024 the DIN SPEC 91504:2024-11 was published as national standard, with the title “Barrier-free charging infrastructure for electric vehicles” <sup>(21)</sup>. In the document are set mainly technical requirements for recharging equipment (including the height of control elements and underride bar, controls and displays, as well as recharging cables) and the environment of the recharging equipment (including the accessibility of the charging equipment, recharging site, location of the

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(19) [https://www.userchi.eu/wp-content/uploads/2023/06/Informe-Ceapat-eletroneras\\_.pdf](https://www.userchi.eu/wp-content/uploads/2023/06/Informe-Ceapat-eletroneras_.pdf)

(20) [https://nationale-leitstelle.de/wp-content/uploads/2023/01/Masterplan-Ladeinfrastruktur-II-der-Bundesregierung\\_Englisch\\_DIN\\_A4\\_barrierefrei.pdf](https://nationale-leitstelle.de/wp-content/uploads/2023/01/Masterplan-Ladeinfrastruktur-II-der-Bundesregierung_Englisch_DIN_A4_barrierefrei.pdf), (15/10/2021)

(21) DIN SPEC 91504 – „Barrierefreie Ladeinfrastruktur für Elektrofahrzeuge“

recharging equipment, underground, collision protection, lighting, obstacle marking, as well as a ground guidance system). In addition to the technical requirements, the guideline defines organisational requirements on support services for the usage of recharging stations such as help desks, call centres, technical support, relay services and emergency services. The guideline emerged from the ELSTA project and was developed together with relevant national and international stakeholders in the charging industry, such as charge point operators, EVSE suppliers and federal offices for accessibility and disabled persons.

In **Austria** RVS 03.07.21 (*Richtlinien und Vorschriften für das Straßenwesen* or guidelines and regulations for road engineering) was published in 2022 for Electric Vehicle Charging Infrastructure in Public Areas. This RVS is to be applied for the design of recharging places (car park including recharging station) on public roads. It is also recommended for the design of recharging bays on private land. It applies to conductive (wired) recharging and is applicable to multi-track electric vehicles (classes M1, N1) <sup>(22)</sup>.

In the **Netherlands**, there is the [Plan “UN Convention - Unlimited participation! Implementation of the UN Convention on the Rights of Persons with Disabilities](#). Although this document focuses primarily on how travellers with a disability can increasingly use public transport independently, some of the recommendations on participation can also relate to the work done under these Guidelines.

**UK** is very advanced in integrating requirements for disabled people in tenders. [BSI published in 2022 accessible electric vehicle charging standard PAS 1899:2022](#) [BSI \(bsigroup.com\)](#) <sup>(23)</sup>. Regarding this standard, the suggestion from the survey respondents is to differentiate requirements on Hardware from Site layout, since site layout is related to other stakeholders and is subject of a discussion between operator and property owners. In some areas these standards can be too prescriptive.

The **French government** has published some prescriptive technical norms regarding the urban & public infrastructure for people with reduced mobility <sup>(24)</sup>. Some inspiration could be taken from it, from measurements to user experience requirements.

In **Sweden**, Region Jämtland Härjedalen and BioFuel Region collaborated within the project SiSL Mellersta Norrland which promotes battery electric cars and the expansion of recharging infrastructure. The objective is to simultaneously increase the number of recharging stations and ensuring that they have an

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(22) <http://www.fsv.at/shop/produktdetail.aspx?IDProdukt=c77e97bb-1de3-473e-89e4-6340334b7713>

(23) <https://www.bsigroup.com/en-GB/about-bsi/media-centre/press-releases/2022/october/accessible-electric-vehicle-charging-standard-to-ensure-accessibility-for-all/>

(24) <https://www.ecologie.gouv.fr/laccessibilite-voirie-et-des-espaces-publics>

inclusive design. The resulting guideline is called [Accessible Charging Stations: Practical advice on designing inclusive charging stations](#).

**In the US** there is the **Americans with Disabilities Act (ADA)**, a civil rights law that prohibits discrimination against individuals with disabilities in all areas of public life, including jobs, schools, transportation, and all public and private places that are open to the general public. The purpose of the law is to make sure that people with disabilities have the same rights and opportunities as everyone else. In addition, there were published Design Recommendations for Accessible Electric Vehicle Charging Stations, US Access Board (2022) (25) and Universal Design principles (26).

**ISO/IEC Guide 71:2014** is a guide developed by the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC) to assist developers on addressing accessibility requirements and recommendations in standards that focus, whether directly or indirectly, on systems (i.e. products, services and built environments) used by people. To assist standards developers to define accessibility requirements and recommendations, it presents a summary of current terminology relating to accessibility, issues to consider in support of accessibility in the standards development process, a set of accessibility goals (used to identify user accessibility needs), descriptions of (and design considerations for) human abilities and characteristics, and strategies for addressing user accessibility needs and design considerations in standards.

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(25) <https://www.access-board.gov/tad/ev/>

(26) <https://www.universaldesign.ie/what-is-universal-design/the-7-principles/the-7-principles.html>

## 5. Existing practices, standards and lessons learnt to facilitate the accessibility of recharging infrastructure in the EU and beyond

In the following sections, we will explore all aspects of recharging points with the overarching goal of enhancing accessibility and usability for everyone.

### 5.1. Main aspects of accessible recharging point

A user's experience of a recharging point will depend on three main aspects of the recharging infrastructure: hardware (design of a recharging point), information about recharging point and interface and finally parking place and surrounding area. Public authorities may have influence on these three key areas of public EV recharging infrastructure. They should be considered carefully when designing and installing EV recharging points, whether on-street, off-street, or at recharging hubs. Failure to do so can create barriers to people being able to charge.

Given the diversity of the population, a universal design approach, which caters to the broadest range of users from the outset, will result in EV recharging infrastructure being as accessible as possible. Collaboration between different sectors, groups, and organisations will be required to deliver EV recharging infrastructure that provides a range of accessibility options to meet user needs.

As stated by the Spanish accessibility criteria for electric vehicle recharging points or EV recharging stations, when planning for accessible and inclusive recharging infrastructure we should take into account the seven principles of universal design: Equitable Use, Flexible, Simple and Intuitive Use, Understandable Information, Error Tolerance, Low Physical Effort, Space and Size for Approach and Use.

### 5.2. Design of the recharging station (technical parameters, standards, payment options)

#### 5.2.1. Component Heights

The height of all recharging components, including the socket, payment terminal, and screen/visual interface, should accommodate users with varying height requirements and mobility aids. It is also important to ensure that, when establishing minimum height requirements for e.g. persons on wheelchairs, this does not result in accessibility constraints for very tall persons. The following table compiles recommendations from different guidelines and standards, from the EU

and beyond, that can be used as guiding principles for when dimensioning accessible recharging stations.

For public charge points situated at sidewalk level without a level access point, where the charge point is accessed from the carriageway level, the design must account for the height of the kerb. This includes considering the height of any mounting base or column component on which the recharging point is installed, ensuring access remains within required ranges. Ideally, lowering the charger's concrete foundation to ground level is recommended to mitigate surface level differences, ensuring that the screen, buttons, and controls are at an appropriate height.

**Table 3 – Recommendations from different guidelines and standards on height requirements for several components of a recharging station**

| Guidelines                       | Recharging point socket-outlet | Charging cable connector handle | Screen/visual interface      | Payment terminal      |
|----------------------------------|--------------------------------|---------------------------------|------------------------------|-----------------------|
| PAS 1899:2022 (UK)               | 800 - 950 mm                   | 800 – 950 mm                    | 800 (bottom) – 1300 (top) mm | 800 – 1000 mm         |
| SiSL Mellersta Norrland (Sweden) | 800 – 1000 mm                  | 800 – 1000 mm                   | 800 – 1000 mm                | 800 – 1000 mm         |
| Zero Emission Vehicles (Ireland) | 900 – 1200 mm                  | 900 – 1200 mm                   | 900 – 1200 mm                | 900 – 1200 mm         |
| DIN SPEC 91504 (Germany)         | 850 – 1050 mm                  | 850 – 1050 mm                   | 850 – 1050 mm                | 850 – 1050 mm         |
| Recharge (Norway)                | 750 – 1200 mm                  | 750 – 1200 mm                   | 750 – 1200 mm                | 750 – 1200 mm         |
| US Access Board (USA)            |                                |                                 | 48" maximum (1220 mm)        | 48" maximum (1220 mm) |

### 5.2.2. Plug Handle

The recharging station's plug handle should be designed with ergonomics in mind, ensuring a secure grip without slipping. This includes considerations like surface finish and handle design. Both tethered and non-tethered cables should allow users to comfortably grip the plug handle with one or two hands, without the need for excessive strength or dexterity. Furthermore, the plug receptacle itself should be ergonomically designed to enhance ease of use.

When possible, recharging point manufacturers should consider installing automated connection devices, especially at fleet charging stations.

Automated connection devices in EV charging stations aim to improve accessibility for persons with disabilities. These devices, such as robotic arms or automated cable management systems, can help users with limited mobility or dexterity to connect and disconnect their vehicles without manual effort.

#### **Some examples in the EU and beyond**

Designability's Design Guidance for Accessible Electric Vehicle Charging recommends a handle that can be held in different ways using one or two hands without relying on strength or dexterity.

According to PAS 1899:2022, the design of a recharging point cable connector grip or handle shall be of a suitable material and ergonomic design to be usable by people including disabled people with strength or dexterity impairments. The diameter of the gripping portion of a recharging point connector grip or handle shall be between 19 mm and 43 mm. Currently, a new version of PAS standard is being prepared in order to update some elements and should be published in March 2026.

### **5.2.3. Cable Weight and Rigidity**

The recharging cable present in DC recharging stations should be as light as technically possible, and its rigidity should be manageable for EV drivers, considering their physical abilities and statures permitted to drive. But cable weight can significantly increase, particularly with high-power output recharging stations providing fast recharging. This persists as a challenge due to the necessity for thicker, heavier cables to enable faster charge times and meet safety requirements. Additionally, these cables may require active cooling, which can make them less flexible and more difficult to manoeuvre into position. For individuals with disabilities who rely on rapid and ultrafast rechargers due to limited access to home or work-based charging, handling heavy cable weights becomes a critical issue. Implementing a cable management system, where applicable, can help solve this issue. Where possible, a recommendation for minimising cable weight is the use of suspended recharging cables which distribute the weight and prevent them from lying in the way on the ground.

Low-powered, on-street rechargers (including lamp-post chargers) represent a cost effective option however adapting them in this respect can also pose challenges. The design of on-street rechargers should address accessibility concerns, both at the individual charge point level and, on a street-wide scale. However in situations where it is not possible to adapt or refurbish all recharging points, implementing 'reasonable accommodation' measures may be necessary.

### **Some examples in the EU and beyond**

PAS 1899:2022 recommends that the force required to lift a cable suspended in the air, with a length of 4 metres, should not exceed 60 Newtons, or the equivalent to lifting around 6 kgs (measure of force).

Designability's Scoping and Discovery Report for Accessible Electric Vehicle Charging suggests a cable weight of approximately three kilograms.

## **5.2.4. Cable Length and placement**

The cable length must be sufficient to reach for all consumers, including persons with disabilities.. It should also be adaptable to various parking positions within the parking bay, ensuring it does not obstruct the surrounding area during recharging, which could pose accessibility challenges for disabled individuals and others.

When not in use, a cable management system, such as overhead cable support, should be implemented where possible to prevent the cable from obstructing roadways and to prevent it from dragging across the ground. Fast rechargers should include cable support to alleviate cable weight. For non-tethered cables (portable and owned by the user), the recharging station should offer a system for supporting and managing the cable, such as the one recommended by Designability.

During recharging and when connected to a vehicle, any unused cable length for tethered public recharging points should be supported either on the ground adjacent to the vehicle or by an appropriate cable management system to avoid presenting a trip hazard. It is important to determine the best method for routing cables beside vehicles while they are recharging, considering force requirements.

### **Some examples in the EU and beyond**

The standard PAS 1899:2022 recommends that the length of the charging cable should not exceed 7,5 metres.

CROW has established guidelines for the application of private charging cables, including the prohibition of loose cables on the sidewalk. If no cable duct is allowed or possible, a cable mat or other solution can only be chosen if this does not result in pedestrians (with or without a walker, wheelchair or stroller) not being able to pass or tripping or slipping.

### 5.2.5. Sockets and Connectors

The sockets at recharging stations should be easily visible and visually distinctive for all drivers (including persons with disabilities for example, colourblind or elderly drivers). They should clearly display the plug type and orientation for user convenience. Tilting the socket upwards at a slight angle can assist taller users in seeing the socket without impeding accessibility for seated users. Additionally, the connection force required for inserting and removing the recharging cable, both from the recharging station and the electric vehicle, should as much as possible be accommodated to users of all abilities while ensuring necessary connection strength.

#### Some examples in the EU and beyond

The PAS 1899:2022 recommends that the connection force, where possible, should be limited to a **maximum of 15 Newtons** (measure of force).

AFIR legislation mandates that charging points for light-duty vehicles should be equipped (at least) with Type 2 connectors for AC connections and Combo 2 connectors for DC connections.

The Irish Electric Vehicle Charging Infrastructure – draft Universal Design Guidelines 2023 socket covers, including a hinge in socket cover door, a push down socket cover, or a slide up socket cover, should be designed with user needs in mind.

Designability's Design Guidance for Accessible Electric Vehicle Charging recommends that where a socket cover is present, users should be able to easily open the socket cover and insert the connector using one hand.

### 5.2.6. Charging Station Visualisation

In order to be clearly perceived visually even in poor weather conditions, recharging stations should be designed in a way that they provide enough visual contrast with their surroundings to ensure that they are easily detectable to road-users, including drivers with disabilities. This should be done by ensuring that adequate indicators are in place, such as light, tactile elements, and different surfaces around the charging station. Clear labelling of key features on the charging station, including the connection type and payment system, should be included. In the EU, in accordance with AFIR Annex II, point 10.2, the 'Identification of vehicles and infrastructures compatibility - Graphical expression for consumer information on EV power supply' shall comply at least with standard EN 17186:2019.

#### Some examples of guidelines in the EU and beyond

DIN SPEC 91504 Guidelines recommend that recharging stations should be designed either to visually contrast strongly with their surroundings or with two 8 cm wide, visually strongly contrasting marking strips at a height of 40 cm to 70 cm and from 1,20 m to 1,60 m above the surface covering or, in the case of bollards (minimum height of 90 cm), at least in the upper third (according to national standard DIN 18040-3:2024-12, 4.4).

The Irish Electric Vehicle Charging Infrastructure – draft Universal Design Guidelines 2023 recommends that the charging station should be Using consistent and contrasting colours for the charging station, its components and surrounding environment can enhance user friendliness. Charging stations can serve as important way finders on large open parking spaces for people with visual impairments. Ideally, charging station locations should be integrated in digital navigation aids.

The Irish Wheelchair Association's Best Practice Access Guidelines recommends any street furniture provided should not intrude into the circulation route and should be clearly detectable by design and contrast against the background. Street furniture which can assist with route identification should be located in a line along or set back into the pavement or approach route.

#### 5.2.7. Other usability aspects

When it comes to complementary recommendations to increase the usability and accessibility of a recharging station, a few aspects should also be considered.

Integrating a walking aid holder into the recharging station offers a convenient place for persons with disabilities to place walking sticks or crutches, enhancing the recharging experience by freeing up their hand(s) to handle the plug. Moreover, the holder can double as a storage solution for excess cable, assisting users in managing non-tethered cables more efficiently.

A footwell or recession, to allow clearance for wheelchair footplate, at the base of the recharging station should be considered.

#### Some examples in the EU and beyond

The Irish Wheelchair Association's Best Practice Access Guidelines recommends a footwell giving 300mm height clearance for wheelchair footplates to the base of the recharging point to allow a person using a wheelchair to have a front-on close approach to the recharging point.

#### 5.2.8. Payment Options

According to the survey respondents, as the table below indicates, contactless payments are the most important to ensure accessibility and inclusivity.

**Table 4 – ranking of the different payment solutions in terms of accessibility and inclusivity, according to survey respondents**

The score was calculated as follows: the first priority was giving the weight of 3 points, the second of 2 points and the third of 1 point.

| Solutions for accessible payment options                                                                                                                                                                                                                                                                                                                | SCORE |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Possibility to use contactless payment options, which are more user-friendly (RFID, bank cards...)                                                                                                                                                                                                                                                      | 42    |
| Defining precise technical dimensions of the payment terminal (for example the height of the equipment like in the American Disabilities Act which now requires that all operable parts of a gas pump be at or below 54 inches above the driving surface, and any new construction must have those parts at no higher than 48 inches above the surface) | 29    |
| Requirements for accessible payment terminal in accordance with Directive 2019/882 (European Accessibility Act)                                                                                                                                                                                                                                         | 27    |
| User interface and dimension as required by EN 301549 (European standard for digital accessibility) and EN 17210 (Accessibility and usability of the built environment - Functional requirements)                                                                                                                                                       | 24    |
| Others (e.g. ISO 15118-20 Plug & Charge)                                                                                                                                                                                                                                                                                                                | 15    |
| Provision of adequate locations for the payment terminals in places where there are multiple rechargers connected to one payment terminal.                                                                                                                                                                                                              | 7     |

To cater to various user needs and simplify the process, recharging stations should offer versatile and user-friendly payment methods, in line with AFIR requirements. Therefore, the integration of **contactless payment** at recharging stations, providing a convenient and efficient method for users to make payments seamlessly represents an important element of making recharging infrastructure accessible for disabled persons and the AFIR requirements contribute to this aspect. Plug & Charge functionalities should also be explored to enhance the customer experience. Whenever possible, contactless card readers should be preferred over those requiring physical card insertion. In cases where physical card insertion is necessary, including a tactile number pad is essential for accessibility. It should be possible to handle these pads with one hand. The payment pads should be easy to use and intuitive, in line with the recharging point screen requirements.

Payment services are covered under the European Accessibility Act and its requirements are of compulsory compliance, as the EAA mandates that payment terminals and other related services must be designed to be accessible. This includes ensuring that user interfaces are usable by people with various

disabilities, providing clear instructions and support, and making sure that all relevant information is available in accessible formats.

### Some examples in the EU and beyond

The Irish Electric Vehicle Charging Infrastructure – draft Universal Design Guidelines 2023 recommends that the implementation of **prepayment options through a digital app or digital interface** could be helpful for disabled persons.

The PAS 1899:2022 suggests that consumers can rely on a familiar payment method across recharging station networks including Radio Frequency Identification (RFID) cards, a contactless payment card and/or a smartphone application.

## 5.3. Parking place and surrounding area (technical and quality requirements)

**UK national disability charities Motability and Designability** have published findings from the user engagement phase of the project to make electric vehicle (EV) recharging infrastructure in the UK accessible to disabled people (2021). Accessible parking with room to manoeuvre around the vehicle is already a key concern for disabled people. In a public recharging environment, the ability to fully open doors, manoeuvre equipment and use lifts and ramps is paramount. In addition, users highlighted that it is important to have flat (non-sloping), smooth (without gravel, grass, mud, cobbles or potholes) parking spaces with level access (without kerbs or steps), or nearby available dropped kerb access onto pedestrian areas and nearby amenities. Here the role of public authorities is crucial

It is also important for disabled people the position of the charge point in relation to the parking space, waiting for the vehicle to be charged (particularly focusing on the availability of accessible facilities nearby), and seeking assistance if they experience difficulties.

According to the project by Region Jämtland Härjedalen and BioFuel Region in Sweden, the most important thing when designing an accessible recharging station is making it easy to access the charger when the vehicle is parked. This means that there should be ideally plenty of space both next to and in front of the vehicle. A good place to begin is to design the recharging station as if it were a family car parking space that is extra-long and with plenty of space marked as a buffer zone on both sides of the vehicle. **However at the same time space efficiency should be taken into consideration.**

When installing slow and fast chargers, it is common to install many chargers in the same location. But sometimes space is scarce and limits, or negates, the possibility of creating extra wide spaces. In such cases, there might be the possibility of still increasing accessibility of the parking space and surrounding area by widening the outer parking spaces. Finally, it is up to the public authorities in the cooperation with landowners to keep right balance between these parking lots and other uses for public space and accordingly introduce requirements in the tendering process.

The buffer zones between and in front of chargers provide good accessibility around the entire vehicle and reduce the risk of a user being blocked in by other vehicles.

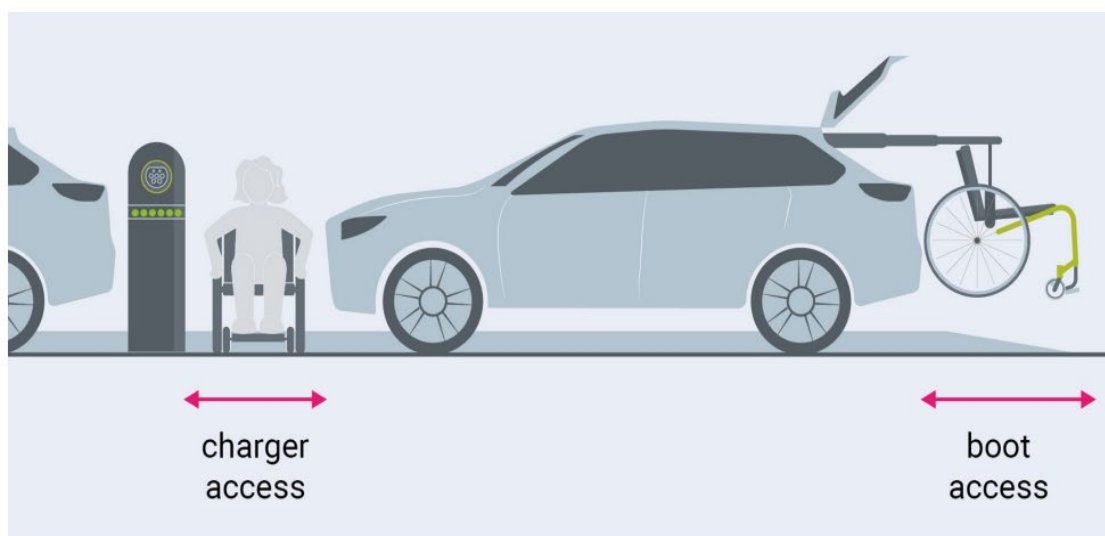
This is a good way to improve accessibility if there is room for buffer zones between each parking space. A zone has been created in front of the parking spaces to make it easier to get around the car and access the charger. A parking space with open space beside it is also often a good way of improving accessibility.

Designability's design guidance recommends for the **space at the sides of a vehicle to be enough to** open driver and passenger's doors fully. This will allow drivers and passengers with reduced mobility to straighten their legs when getting in and out, to have enough space to help passengers (adults or children) in and out, and to have enough space to move along the sides of the vehicle.

Some adapted vehicles need space along the side so that the user can get in and out using ramps or lifts. Some users will need to open doors on both sides of the vehicle on the same occasion. For example, a disabled driver may need to get out of the driver's door and then have enough space to manoeuvre around to help a child out of a passenger door on the other side of the vehicle.

Regarding the space at the front and back of a vehicle, it advises ideally to provide protected space at the front and back of the vehicle at the same time. For example, a user may need to park their vehicle with the front recharging socket close to a recharging point leaving enough space to access the recharging unit and still have enough space to get their wheelchair out of the boot while being protected from moving vehicles. It is important that protected space around the vehicle is clearly marked so that drivers do not park in it.

**Figure 6 – Designability recommendations for space around a vehicle and an EV charger**



Source: Designability

Designability recommendations for space around a vehicle and an EV charger .

### 5.3.1. Parking Space Dimensions

Ideally, parking spaces should be designed to facilitate accessibility for all users, allowing them to easily enter and exit their vehicles, manoeuvre safely around the vehicle, access the recharging station, and park comfortably. Perpendicular and parallel parking are the most common types, and their designs can serve as models for other parking configurations.

While it is not feasible to universally stipulate a minimum level of space dimensions for existing parking spaces stations (in particular in urban areas with narrow streets or historical city centres, where space is already scarce), the following table compiles recommendations from different guidelines and standards, from the EU and beyond, that can be used as guiding principles for the parking space dimensions for accessible recharging stations.

Where a minimum sufficient width cannot be guaranteed, level access along the full length of the pavement should be provided.

**Table 5 - Recommendations from different guidelines and standards on the parking space dimensions of an accessible recharging station**

| Guidelines                                            | Length of parking space            | Width of parking space             | Access zone surrounding the parking place |
|-------------------------------------------------------|------------------------------------|------------------------------------|-------------------------------------------|
| PAS 1899:2022 (UK)                                    |                                    |                                    | 1200 mm                                   |
| SiSL Mellersta Norrland (Sweden)                      |                                    |                                    | 900 mm                                    |
| Zero Emission Vehicles (Ireland)                      | 4800 mm                            | 2400 mm                            | 1200 mm – 1500 mm                         |
| DIN SPEC 91504 (Germany)<br>Longitudinal charging bay | 5800 mm                            | 2000mm                             | 1500 mm                                   |
| DIN SPEC 91504 (Germany)<br>Vertical charging bay     | 5200 mm                            | 3650 mm                            | 1500 mm                                   |
| DIN SPEC 91504 (Germany)<br>Drive-through solution    | 5800 mm                            | 2000mm                             | 1500 mm                                   |
| Recharge (Norway)                                     | 6000 mm<br>(including access zone) | 4500 mm<br>(including access zone) |                                           |
| CEAPAT (Spain)                                        | 5000 mm                            | 2200 mm                            | 1200 mm                                   |
| US Access Board (USA)                                 | 20' (6096 mm)                      | 11' (3356 mm)                      | 5' (1524 mm)                              |

### Some examples in the EU and beyond

The PAS 1899:2022 recommends a minimum barrier of 1,2 metres between the vehicle and the recharging station. A minimum space of 1,8 metres would enable a wheelchair to turn through a 180° angle.

The Irish Wheelchair Association suggests all recharging stations for electric vehicles should include a standard designated accessible parking bay with level access to the use of the recharging equipment and including a 1.2 metres, but preferably a 1.5 metres, access zone between the vehicle and the recharging station, to allow sufficient space for a person who is a wheelchair user to manoeuvre and gain access to the recharging equipment and to the vehicle connection point.

Motability's Scoping and Discovery Report: "Accessible EV Charging" recommends the site design should consider varying vehicle socket location, therefore providing ample space for different parking orientations to safely manoeuvre and charge the vehicle.

### 5.3.2. Ground Surface Type

Public recharging stations should be installed on a flat, stable, and non-sloping surface with adequate grip to ensure safety and ease of use. Different surfaces surrounding the recharging station should be used to indicate orientation and the presence of obstacles. An appropriate surface gradient should be provided beneath the recharging station, and proper water drainage facilities should be in place to prevent the formation of puddles. Level and smooth ground is crucial, particularly for individuals with mobility issues, as sloping or uneven surfaces like gravel, potholes, or mud can pose significant challenges and increase the risk of tripping or falling. It is essential to ensure that users do not need to navigate muddy, grassy, or uneven terrain to access the recharging unit once they have arrived.

#### Some examples in the EU and beyond

The National Disability Authority's (NDA's <https://nda.ie/>) Building for Everyone: A Universal Design Approach recommends a dry friction coefficient (surface friction) between 35 and 45 (PTV).

DIN SPEC 91504 Guidelines defines that the access areas around the recharging station are firm, even, can be rolled over with low vibration, and are slip-resistant for barrier-free use (according to national standard DIN 18040-3:2024-12, 4.4).

The PAS 1899:2022 recommends that the gradient should not be steeper than 1:50.

According to SISL Mellersta Norrland recommendations, the ground should not slope more than 1:50.

### 5.3.3. Ground Surface Height Differences

Whenever feasible, it is preferable to position recharging stations on the same level as the roadway to facilitate easy access. However, in cases where height differences are unavoidable, disabled access features such as kerb drops and ramps, built in compliance with existing regulations, should be provided. It is important to ensure that drainage infrastructure does not obstruct access to the recharging station.

### **Some examples in the EU and beyond**

The Irish Wheelchair Association's Best Practice Access Guidelines recommends installing a kerb with a slip resistant surface with a minimum width of 1.2 metres and a gradient no steeper than 1:12.

DIN SPEC 91504 Guidelines defines a maximum longitudinal slope of 3 % and transverse slope of 2 % (according to national standard DIN 18040-3:2024-12, 4.3a).

The SISL Mellersta Norrland manual suggests it is important to remove as many obstacles as possible, such as different surface levels.

The PAS 1899:2022 recommends a dropped kerb within a maximum of 20 metres distance of the recharging station.

### **5.3.4. Removing Obstacles**

Wherever possible, obstacles surrounding the recharging station should be eliminated. It is crucial to recognize that any barrier impeding a wheelchair user from reaching within 25 cm of an object significantly limits their ability to access the recharging point. Therefore, ensuring that bollards and plinths do not unnecessarily increase the reach required is paramount.

Additionally, other obstacles such as wheel stops, safety barriers, drainage infrastructure, or kerbs should be appropriately positioned to allow sufficient access to the recharging station and avoid obstructing other road users.

Unmanaged cables, whether tethered or untethered, pose a trip hazard and should not obstruct other road users, including those on footpaths and cycling lanes. It is important to consider that some wheelchairs and scooters have large footplates or components that may limit a person's reach.

Recharging point operators have noted that additional one-off costs may arise for customizing the design of on-street charge points for disabled customers. This includes expenses related to installing dropped kerbs, making adjustments to the built environment (such as adding signage), or enhancing the accessibility of charge point software interfaces.

### **Some examples in the EU and beyond**

The Irish Wheelchair Association's Best Practice Access Guidelines recommends that street and pavement clutter should be avoided.

The PAS 1899:2022 suggests that there should be no obstacles in front of the recharging station, so that the reach distance is 0 metres.

CEAPAT in its report on “Accessibility criteria for electric vehicle recharging points or recharging stations” recommends that accessible charging points will guarantee access from the transfer area to the accessible pedestrian itinerary, autonomously, free of obstacles and safely.

The project SISL Mellersta Norrland in Sweden recommends that Collision barriers must keep cars away from the recharging station in case of an accident, but they may not impede accessibility and stop wheelchairs users. Collision barriers are sometimes placed too close together or consist of an arch in front of the charger, which reduces accessibility (place them at least 90 cm, preferably 120 cm apart to provide good accessibility). Sometimes recharging handles and outlets, screens, buttons and card/RFID readers are placed on different sides of the charger. This makes it particularly important that the charger purchaser be aware of how weather protection and collision barriers will be positioned.

### 5.3.5. Access to Amenities

Recharging stations located in highways should be positioned, where possible, in close proximity to any nearby amenities, such as shops, toilets, cafes, or other services, whenever feasible. Universally designed public seating may be incorporated in the vicinity of the recharging station, provided it does not obstruct individuals. When possible, already existing infrastructure that facilitates accessibility should be considered when deploying recharging points/poles/stations.

#### **Some examples in the EU and beyond**

The PAS 1899:2022 recommends building the recharging station as near as possible to the amenities.

US Access Board Accessibility Standards recommend that an EV recharging station must connect to an accessible route that leads to an accessible entrance of the building or facilities on the same site. Additionally, the accessible EV rechargers should be on the shortest accessible route to the accessible entrance relative to other rechargers at the same recharging station.

### 5.3.6. Weather Protection

Where possible, adequate overhead weather protection should be placed around recharging stations located on highways and along access routes from the recharging stations to nearby amenities or facilities to minimise the impact of adverse weather conditions on users.

### **Some examples in the EU and beyond**

The US Access Board Accessibility Standards recommends installing weather protection, security and adequate lighting at parking spaces.

The project SISL Mellersta Norrland recommends that weather protection should also be designed so that it does not inhibit access to recharging contacts, screens, buttons and card readers. Note that in some cases, this equipment is placed on multiple sides of the charger.

### **5.3.7. Safety Considerations**

Ensuring a safe and secure recharging environment is essential for all users. In recharging stations located in highways, clearly visible security cameras with a direct view of the recharging stations can help deter crime. Sightlines to other users and nearby amenities should also be considered and recharging infrastructure should be placed near facilities. Public authorities may require that, when possible, recharging should be in the most highly trafficked sections of a carpark to facilitate passive surveillance and a feeling of safety for users. Additionally, when placing recharging infrastructure, public authorities and other relevant stakeholders should consider aspects of safety to minimise the occurrence of accidents, in particular for vulnerable users.

### **Some examples in the EU and beyond**

The PAS 1899:2022 recommends security cameras should be provided in the vicinity of public recharging stations and particularly in the vicinity of public recharging station located at designated accessible parking bays. Security cameras should be visible such that it is obvious that an area containing public recharging station is protected by the security cameras and should be maintained in good working order for monitoring and maintenance.

### **5.3.8. Lighting**

Adequate and consistent lighting following relevant standards throughout different parts of the day is crucial. This will allow all users to safely navigate the environment surrounding the charger, use the equipment, and access the routes between the public recharging stations and nearby amenities.

### **Some examples in the EU and beyond**

The Irish Wheelchair Association suggests access routes to car and multi-purpose vehicle bays should be adequately illuminated with a recommended 150 lux level.

The PAS 1899:2022 recommends lighting should be provided in the environment surrounding public recharging station, to enable safe use of the recharging station, as well as routes from the public recharging station to any nearby venues or amenities.

According to Recharge guidelines for accessibility, the recharging station screen must be illuminated with at least 500 lux, and the area around the charger should have approximately 200 lux.

A recital to the AFIR legislation also suggests ensuring adequate lighting at recharging stations.

### 5.3.9. Recharging Station Placement

The positioning of the recharging station should prioritize unobstructed sightlines for other road users and safe operation. It should also be adaptable to various vehicle types and minimize obstruction in its surroundings. To offer users flexibility in vehicle orientation, recharging stations should be placed in variable positions to allow for multiple parking orientations.

For on-street locations, recharging stations should be ideally situated on kerb extensions to prevent obstruction of footpaths. En-route fast recharging stations may consider a side-on layout akin to existing petrol stations to accommodate longer vehicles. If a recharging station serves multiple recharging bays, it should be centrally located between those bays.

Furthermore, it is important to note that while foliage can enhance the aesthetics of car parks, untrimmed shrubs and bushes may impede recharging, particularly in wet weather, making it unpleasant or impossible for some users.

#### **Some examples in the EU and beyond**

According to the project SISL Mellersta Norrland, it is important to consider suitable locations in the car park if possible, avoiding installing the recharging station in the car park's most attractive spaces. This reduces the risk of the space being used by drivers who do not need to charge. However, a location far from the entrance or service point reduces accessibility, so there is a need to balance these conflicting needs. Additional points to take into account, especially in Nordic countries is to select a location that is cleared of snow in the winter.

The Swedish Accessible Recharging Station manual suggests positioning the recharging station like a traditional petrol station which makes it easy to exit after recharging and makes it easier for longer vehicles and vehicles with trailers to charge.

The PAS 1899:2022 recommends that the placement of a public recharging station, in combination with other existing furniture, should not further compromise the accessibility of the surrounding environment.

## 5.4. Information/data about recharging points and interface

### 5.4.1. Recharging Station Digital Interface

In the reference documents assessed by the Taskforce, possible options to increase the accessibility of the digital interface include the following:

- **A high contrast interface.** For people living with visual impairment who are allowed to drive, it is highly important that high contrast colour themes are prioritized over modern design choices.
- **Brightness** in different light condition. Recharging station should be illuminated for all drivers in a way that all controls can be recognised and operated even in the dark.
- **Using of fonts, text and icons.**
  - When choosing a font, it is important to avoid the ones that might cause recognition difficulties for people with visual impairment and/or dyslexia because the font has little amount of aperture. This means that the chosen display font must have white space within the letters to avoid the lack of readability.
  - Using icons and visuals can also improve the user experience of people living with visual impairments and/or dyslexia.
- **Angle of the screen.** The angle of the screen at the recharging station should ideally be adjustable to improve readability from a seated or fully upright position and to mitigate sun glare.
- **Touch-sensitive** screen for ease of use. This would naturally be applicable only in specific situations, as some equipments have buttons and controls, but not a screen.
- **Buttons and controls** of a recharging station tactilely and visually distinguishable, including **night-time functionality** (according to the *Einfach laden ohne Hindernisse*).

In addition, a digital display should provide clear and simple instructions at every step of the recharging process. Instructions should be in layperson's terms and language options should be available. Speech-based navigation and screen reader mode could be eventually integrated to aid the user with disabilities, who are allowed to drive, through the recharging process.

#### **Some examples in the EU and beyond**

The PAS 1899:2022 recommends that the screen interface should be tilted at an angle between 0 degrees and 20 degrees upwards from the vertical plane towards the user of the recharging station.

Recharge recommends that the screen should have an angle between 55 - 70 degrees off horizontal position.

On the basis of the Irish Electric Vehicle Charging Infrastructure – draft Universal Design Guidelines 2023, for recharging stations, it is crucial that the recharging point and the screen do not sit so high that they are above shoulder height of a person in a wheelchair.

### **5.4.2. Mobile Applications**

Mobile applications are widely used to facilitate the recharging process and enhance user experience by offering access to information before, during, and after a recharging session. Providing remote feedback during the recharging process could be one options in the future to enable users to monitor the recharging session status and manage the recharging process without the need to constantly exit or return to their vehicle.

In order for a driver to charge the vehicle properly, both the mobile application and the access to the recharging point digital service should meet accessibility criteria, in accordance with existing legislation and regulations.

### **5.4.3. Recharging Session Feedback**

During the recharging process, providing feedback is crucial to communicate the stage of the recharging process to users. This includes confirming successful actions such as plugging in the connector, readiness to unplug, and verifying payment completion. In the event of a recharging failure, clear instructions for next steps, such as locating the next available recharging station, should be provided.

Consistency in light cues across charge point operators is important, ensuring visibility even in daylight. It is recommended to use distinct colours as provided by the industry-followed protocols today.

For inclusivity, audio cues can be added to complement visual cues, especially for drivers who may be colour-blind. However, vibration cues should be avoided to prevent confusion.

#### **Some examples in the EU and beyond**

The PAS 1899:2022 points to research that recommends the following colours for the different operational stages: green indicates the device is operational; blue indicates the device is in use; red indicates the device is non-operational; and blue flashing indicates recharging is finished.

#### **5.4.4. Provision of data on recharging infrastructure**

According to the experience gathered by the UK Motability and Designability, the EV sector was frequently described as complex and confusing, with users typically feeling like they had to carry out a significant amount of research when it comes to recharging an EV. For disabled users, it can be more difficult because of the need to determine charge point accessibility ahead of arrival and the accessibility of local amenities. The lack of reliable, accessible public recharging infrastructure and signposting on accessibility left many feeling that they had no choice but to build significant contingency into their travel plans (e.g. keeping their battery well-charged or planning several possible charge points for a single recharging occasion). The use of the vehicle battery to operate hoists, lifts, air conditioning and heating (for medical or comfort reasons) could also have a disproportionate effect on the available travel range for disabled people.

Therefore, information about recharging points is important to help people choose a suitable one before they set off. Without easy access to this information disabled people can spend unnecessary time and effort looking for suitable recharging points.

In the UK there are currently no known eMSPs providing accessibility data as part of their service. The most useful existing data for users to judge accessibility of existing charge points is crowd-sourced photographs of charge point installations on eMSP platforms. However, coverage is entirely dependent on EV user submissions and not all photographs will be useful to understand accessibility.

Information about location and accessibility could easily be integrated into already existing applications of local and regional municipalities or transport operators.

Providing an example is the Floya application in Brussels launched by the local public-transport operator company STIB-MIVB and Brussels Mobility. This app currently includes both private and public transportation services (e.g. Public Transport timetable; Lime/Villo availability information and pricing; etc.). Implementing solutions like this for the benefit of the people requires a strong public-private cooperation both on a local and federal level. Having transparent information about recharging points and reliable up-to-date data are the crucial first step of informing about accessible recharging points in the EU (according to the delegated/implementing acts of AFIR). The basis for some recommendations on this topic is presented in Chapter 6.2 of this document.

The decree of the Spanish Ministry of Ecological Transition from May 2023 requires OCPI 2.1 protocol to include the “accessibility” field (“Accessibility for people with reduced mobility”: the recharging point is accessible for people with reduced mobility (yes/no/not available)). Recently CENEX in the UK made a survey and is currently finalizing a report which would propose to include to the data to be provided by CPOs more details for example the measures of parking space at a charger..

A key issue is to ensure that the detail provided in the accessibility data specification is appropriate. There is a balance to be found; ensuring that sufficient data is provided to meet the needs of end-users whilst ensuring that the data is not unnecessarily onerous for CPOs to gather and maintain. There is a wide spectrum with regards to the level of detail for the data specification. At the least detailed end of this spectrum, CPOs could simply share whether a recharging point installation is accessible for disabled people (according to the requirements defined by the AFIR delegated and implementing acts). At the other end, if possible, data could be shared on every single aspect of accessibility of the charge point, its installation, the wider built environment and the facilities associated with the charge point’s location.

### **Some examples in the EU and beyond**

The PAS 1899:2022 suggests that the provision of data and information regarding the locations and characteristics of public recharging stations can assist drivers to plan their journeys in advance.

The AFIR legislation (Article 20 on Data provision) requires operators to ensure the availability of static and dynamic data relating to electric vehicles recharging infrastructure and allow access to this data to users through the Member State’s National Access Point, at no cost. Recent Delegated and Implementing

Acts<sup>27,28,29</sup>, in particular Commission Implementing Regulation (EU) 2025/655, further specifies a comprehensive list of required data types to be made available, including data on the number of parking spaces for people with disabilities. To fulfil these requirements, recharging infrastructure should count with digital systems capable of gathering and transmitting these data, while providing user-friendly and interoperable interfaces to end users in the official language(s) of the Member States and in English.

Having a formal grading system of accessibility at recharging points could help users quickly assess their suitability. Alternatively, a more open review system could allow users to comment on accessibility of recharging points.

These concerns should also be extended to existing websites, in which they could eventually include accessibility symbols for the inclusive EV recharging stations, for example <https://electrify.brussels/en/where-to-charge>.

#### 5.4.5. Wireless recharging, inductive recharging or wireless power transfer

This technology could be an excellent solution for accessibility of recharging infrastructure for disabled persons however currently it is still not commercially feasible

Wireless recharging for EVs offers several advantages, making it a potentially transformative solution. Here are some reasons why:

1. **Reduced Physical Effort:** For individuals with mobility challenges, traditional charging stations often require getting out of the car, navigating on the charger interface, and physically connecting cables. Wireless charging eliminates this step, allowing people to charge their EVs without leaving their vehicle, minimizing physical strain.
2. **Increased Independence:** People with disabilities often rely on assistance to manage physical tasks. Wireless recharging promotes independence, as they won't need help to plug in a charging cable, managing all the process through an application, offering a more seamless experience.
3. **Convenience:** Wireless charging can be integrated into private (home or work) and public recharging spaces, making it possible to charge the vehicle simply by parking over a wireless charging pad. This eliminates the

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(27) <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32025R0655>

(28) [https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/14362-Data-on-alternative-fuels-infrastructure-common-technical-requirements-for-a-common-application-interface\\_en](https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/14362-Data-on-alternative-fuels-infrastructure-common-technical-requirements-for-a-common-application-interface_en)

(29) [https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/14345-Data-on-alternative-fuels-infrastructure-technical-specifications-format-frequency-and-quality\\_en](https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/14345-Data-on-alternative-fuels-infrastructure-technical-specifications-format-frequency-and-quality_en)

need for precise handling of plugs, particularly in poorly lit or crowded environments, which could be also challenging for people with disabilities.

#### Some examples in the EU and beyond

Articles and announcements published in 2023 suggested that vehicle OEMs were getting ready to offer wireless options for private recharging <sup>(30)</sup>. There are several pilots with taxis and buses in Norway, UK and Germany. Regarding public recharging, ISO 15118-20 also considers WPT (park and charge) <https://www.userchi.eu/products/inducar-inductive-charging-for-e-cars/> ).

### 5.5. Distribution of accessible recharging stations in urban areas and along TENT-T corridors (percentage, area based and location indicators)

One of the main drivers behind this work was the suggestion by the Task Force to recommend the establishment of a fixed minimum percentage of accessible recharging places. However, after the several Task Force meetings and exchanges the expert group agreed that such an approach might not be enough to ensure accessible recharging infrastructure where it is most needed (for example, one region might be complying with a required minimum % of accessible recharging stations, but there might not be an even geographical distribution, thus leaving areas underserved. Therefore, a different approach is needed, combining both a % of recharging stations that should be inclusive and meet accessibility requirements (both in terms of the hardware and the surrounding environment) and other location-based criteria in urban areas. This was also clearly stated by the survey respondents.

This combined approach of percentage base and area-based is also required by disabled EV drivers. “The ‘Enable Final report’ realised for UK Power Networks in 2022 includes the results of the survey with disabled customers. On the proximity to charge points, just under half of respondents indicated that a residential on-street charge point would need to be a maximum of 50 metres from home.

One of the survey respondents highlighted that a **% based minimal number of prioritized parking bays depending on the type of locations** could be implemented, but establishing a fixed percentage does not take into account the availability of recharging point locations vs the total dimension of the parking lots (meaning that 5% of 10 spaces is not equal to 5% of 200 spaces in nominal

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(30) <https://www.motorbiscuit.com/bmw-launches-wireless-charging-pilot-program/>, <https://electrek.co/2023/03/02/tesla-teases-wireless-home-charging-station/>, <https://www.electrive.com/2023/02/16/stellantis-has-hevo-working-on-wireless-charging/>

terms). From EU/regional and/or national perspective, at least the recharging point type (urban node, rural, highway, public parking, regular parking lots) should be differentiated in order to give guidance to operators in multiple recharging operation segments. Respondents also mentioned that flexibility to operators and/or landowners should be given, when defining location specific ratios based on the parking demands, peak times and number of recharging points from one or multiple operators.

Although it is quite difficult to establish a percentage that is adequate for all situations, some Member-States have already started working on addressing this issue.

When it comes to the location based indicator, this should be defined by the competent authority based on area, a set of clear criteria (e.g. at supermarkets, hospitals/medical centres/ other urban locations, fast recharging hubs), but also based on the specific needs and user's requests.

### **Some examples in the EU and beyond**

UK standard PAS 1899:2022 refers BS 8300-1:2018 *Design of an accessible and inclusive built environment. External Environment. Code of practice*, which recommends a minimum provision of designated accessible parking spaces and enlarged parking spaces dependent on the type of car park, with an approximate 5% of parking bays recommended to be designated accessible parking bays and 5% of parking bays recommended to be enlarged space.

Recently published in France is the “*Arrêté du 27 octobre 2023 relatif à l'accessibilité des places de stationnement en voirie communale équipées ou pré-équipées de dispositif de recharge pris en application de l'article L. 2224-37 du code général des collectivités territoriales*”, which among other aspects, rates for spaces equipped with recharging stations (or pre-equipped) considering also accessible recharging places, depending on the number of parking places.

These include degressive rates based on the number of recharging stations in the area, and progressive rates over time, as well as relatively flexible application of target rates: starting at 10% (above 200 parking places), then 15% and beyond, depending on the number of parking places (up to 35% for 10 and less parking places, with a flat rate of 1 for 5 or less parking places). This includes accessible sites already developed. It also ensures minimum requirements for longer parking places spaces adapted for persons with disabilities (7 to 9 m), as well as annual communication requirements to the Intercommunal Accessibility Commission.

CEAPAT (the State Reference Centre for Personal Autonomy and Technical Aids in Spain) in its report on “Accessibility criteria for electric vehicle recharging points or recharging stations” recommends that for in service (“petrol”) stations, parkings, or in any place where there is more than one recharging point, a number

of accessible recharging points proportional to the total available recharging points will be reserved. In the case of infrastructures with a single recharging point, this will be accessible.

Recharging point operators indicate that the low utilisation of on-street charge points reserved for disabled customers limits the commercial viability for the widespread installation of accessible recharging points. **Financial support will likely be necessary to facilitate the broader deployment of on-street charge points for disabled users. This support is essential to address the financial hurdles that charge point operators and local authorities have encountered in relation to existing funding support mechanisms.**

According to the “**Irish Electric Vehicle Charging Infrastructure – draft Universal Design Guidelines 2023**” the ambition is for all recharging infrastructure in Ireland to be universally accessible for all. Disabled persons should be able to access any recharging station in the country. However, where this may not be feasible, due to real-world limitations, other provisions should be made to accommodate disabled person’s parking card holders. This could include providing designated disabled parking recharging spaces using the same regulations as in place for regular parking spaces.

## 5.6. The importance of standards

In the UK, in October 2022, an accessible electric vehicle (EV) recharging standard was launched to support the rollout of accessible and inclusive public recharging infrastructure. The standard, known as PAS 1899:2022, has been produced by the British Standards Institution (BSI) and co-sponsored by Motability, the national disability charity, and the UK Government’s Office for Zero Emission Vehicles (OZEV). PAS 1899 sets out the minimum specification for an accessible public charge point, by providing designers, procurers, and installers with detailed requirements and recommendations on how to provide accessible public recharging infrastructure for EVs.

On the document ‘Electric vehicle adoption for disabled consumers’, Energy Saving Trust highlights the lack of common design standards as an issue for recharging point operators, and the fundamental reason for their lack of consideration in ensuring accessibility in the UK. Positively, many recharging point operators in the UK welcomed the forthcoming accessible recharging standards from PAS 1899:2022, saying it would help each of them align designs to meet the needs of more consumers. Although recharging point operators claimed that they did consider reach zones and appropriate heights for wheelchair users, this did not always result in an accessible solution. As an example, it refers to the fact that the connector may be at the correct height to the base of the recharging point, however, this doesn’t include the additional

height of the kerb or the fact that a wheelchair user cannot physically get close enough to the charge point due to a crash barrier. This in turn highlights the importance of specific standards for recharging points, which consider all the possible use cases. It is important to recognise that it is not just a physical accessibility problem, but also a digital one. Consumers often remarked at the complexity of the different digital platforms. Complex payment systems, poor user interface designs, illegible displays were all referenced as issues.

As a key aspect highlighted, the document indicates that firstly, the standards need to include all digital and physical aspects of recharging points and their environment, including signage. This will ensure that all interactions can be made more accessible, not just the physical elements.

As part of the work developed by the STF members on the creation of these Guidelines, a specific group of experts analysed a series of standards relating to accessibility, namely EN 17210:2021, EN 301549:2021 and EN 17161:2019 from the point of view of the content and comprehensiveness, compatibility/incompatibility, consistency/inconsistency between them. The work developed by this taskforce will continue after this document publication, including joint work on possible future standards with the European Standardisation Organisations (ESOs).

The STF members involved in this work highlighted some aspects that are summarised below, with more detailed assessment in the following subchapters:

1. An overarching specific standard for accessibility of public recharging infrastructure is needed since none of the related existing standards cover adequately the full spectrum of requirements and concrete needs of recharging infrastructure specifications and recharging station layouts in support of point 1.18 of Annex II “Technical specifications” of Regulation (EU) 2023/1804 on the deployment of alternative fuels infrastructure (AFIR).
2. EN 17210:2021 would be the best positioned to include minimal technical requirements for recharging infrastructure and therefore to refer to by the new Accessibility standard.
3. EN 301549:2021 covers the accessibility of payment terminals and self-service terminal and as such remains covering elements of the recharging infrastructure.
4. All related existing standards do not cover the full spectrum of use cases (slow chargers vs fast chargers, public vs private). To what extent the minimal requirements apply for what use cases exactly, could be specified in a separate Accessibility standard.

### 5.6.1. Assessment of EN 17210:2021 “Accessibility and usability of the built environment”

The core mind-set and value of the EN 17210:2021 standard ‘Accessibility and usability of the built environment - Functional requirements’ is to describe basic, common minimum functional requirements and recommendations, applicable across the full spectrum of the built environment, for an accessible and usable built environment, following the Design for All/Universal Design principles which will facilitate equitable and safe use for a wide range of users, including persons with disabilities. The functional accessibility and usability requirements and recommendations described in this standard are relevant to the design, construction, refurbishment, adaptation, and maintenance of built environments including outdoor pedestrian and urban areas.

The elements in the standard EN 17210:2021 that are relevant for accessibility of recharging infrastructure for drivers with disabilities would be the following:

- **Accessible and usable pedestrian areas.** It could correspond to the surrounding parking area at a recharging infrastructure.
- **Accessible and usable equipment and facilities.** It could refer to the height and accessibility of a recharging point/hardware and the accessibility of a screen on a recharging point; also, the chapter on ICT user interfaces in the standard could be relevant here but only if amended to include recharging infrastructure. It encourages the use of high contrast displays, buttons, and recharging plugs that are placed at an appropriate height and within easy reach for wheelchair users and others with limited mobility. However, concerning the interface, it is very much focused on ICT user interface, which is only relevant for the chargers with a display which are mostly DC recharging stations.
- **Accessible and usable information via multiple senses:** It can refer to how information is presented on a recharging point hardware via screens.
- **Wayfinding: orientation and navigation, information.** It can include tactile and audible information with hearing enhancement **and visual contrast.** It can refer to the information allowing to find a recharging infrastructure, for example on the station and to the way in which the information is shown on a screen of a recharging point.
- **Placement of signs in the outdoor environment:** It can refer to height of signs, font and size of lettering, visual contrast, illumination of signs etc. (also to any signs related to recharging infrastructure).
- **Access in the outdoor environment/accessible routes** (it can refer to a route connecting a recharging point with facilities on the station).

- **Chapter on bollards:** lit can refer to the bollards placed around a recharging point for its protection.
- **Design of designated accessible parking space.** It can correspond to requirements regarding space around an electric vehicle and a recharging point; there is even one statement concerning it:

*“Where an accessible parking space has a recharging station for electric cars, the additional requirements for reaching and using according to 20.10 Service stations shall be provided”.*

- Minimum length: 5.486 meter
  - Minimum width: 2.743 meter
  - For a 2-space recharging station: min. dimensions 7.200-meter length x 5.400-meter width
- 
- **Access from parking space to an adjacent higher pedestrian path,** for example the issue of kerbs and surface/gradient. It can concern the transfer between a parking space at a recharging point and a pedestrian path.
  - **Lighting:** It can concern recharging station.
  - **Weather Protection:** It can concern recharging station.
  - **Access to amenities and removing surrounding obstacles** (it can concern recharging station)

Below are some missing elements in the standard EN 17210:2021 regarding the accessibility of recharging infrastructure:

- Avoid guidance in wording as: “User interface shall be easy to reach and operate”, “Information in simple wording” or “User interfaces of self-service fuel pumps shall be easy to understand”. They are very generic and open for interpretation. Although difficult, it might be good to specify it a bit more.
- Specification of the minimal requirements on core elements of recharging infrastructure hardware applicable to the various forms of recharging infrastructure types (AC/DC, public/building/office recharging) like plug handle, payment functionality and display (if applicable).
- Standard and minimal requirements for recharging infrastructure categories: DC fast recharging, AC kerb side recharging (dimensions in order to be accessible to people with reduced mobility, especially for wheelchair users), accessibility to AC private recharging shall be specified (including dimensions of surroundings and excluding

obstacles). All operable parts (payments, display, plug handle) minimal and maximal dimensions shall be defined in order to create some flexibility for recharging infrastructure suppliers and operators.

- Commonly used and agreed terms and icons. Currently all kind of different terms and icons are used for the same information. This is not supporting accessibility.
- The technical specifications for a support button to get assistance needed. Adding harmonised requirements on this toll would provide trust to use the recharging infrastructure knowing that you can always get support.

**In conclusion**, the standard EN 17210:2021 includes many requirements that are relevant to the accessibility of recharging infrastructure however some of them would need to be updated and be made more specific. At the same time, a few important elements, mainly on the minimal core requirements for recharging infrastructure elements are still missing as they do not always relate to the built environment. Those minimal requirements may be specified per element of recharging infrastructure hardware, and moreover, be specified per recharging infrastructure type (AC/DC/Public/private). There would need to be also added standards regarding refuelling hydrogen stations.

This role is played currently by the technical requirements to make built environment accessible which exists as a separate document that was developed by standardization bodies together with [CEN/TR 17621:2021 Accessibility and usability of the built environment - Technical performance criteria and specifications](#). However, unlike the EN standard, which is recognized and used by all national standardization bodies (it is a national recognized standard once it is adopted), the TR is an informative support document. In this context, it can be used only as reference for the future development of a new standard with technical requirements or technical specifications.

It is important to note that currently EN1721 is under revision in accordance with M587 and part of the work include incorporating elements of the TR 17621 into EN 1721.

#### 5.6.2. Assessment of EN 301549:2021 “Accessibility requirements for ICT) products and services in the European Union

[EN 301549:2021](#) is the European standard of Accessibility requirements for Information and Communications Technology (ICT) products and services in the European Union.

It regards user interface and others, specifies requirements for information and communications technology to be accessible for people with disabilities. It provides some high-level dimensional guidance for operable parts of stationary ICT products like recharging stations however it does not comprehensively define all dimensional aspects.

It covers accessibility requirements for computer hardware like desktops, laptops, tablets, smartphones, etc. Hardware must have operable parts that can be used by persons with limited dexterity, strength or reach. This includes input controls, connectors, locking mechanisms, etc. It also must contain provisions related to payment terminals.

The elements in the standard EN 301549:2021 that are relevant for a recharging infrastructure for people with disabilities are the following:

- The standard EN 301549:2021 is interesting for recharging infrastructure as it is not only focusing on the physical recharging stations (the hardware) but also on usage via web or mobile app.
- Reach requirements for operable parts and displays on IT products (HW) which can refer to a screen at the recharging station:
  - Operable parts: 1.22 meter from ground
  - Displays: 1.40 meter from ground

The identified missing elements in the standard EN 301549:2021 regarding the accessibility of recharging infrastructure:

- Recharging infrastructure requirements (AC and DC) are not specifically mentioned, neither on-site layout elements related to accessibility.

In conclusion, the standard EN 301549:2021 contains certain elements which correspond to digital aspects of recharging infrastructure, and those elements are also subject to a broader user experience context. However, in order to fully respond to the point 1.18 of Annex II “Technical specifications” of Regulation (EU) 2023/1804 on AFIR, it would need to be adjusted where relevant and complemented by specific minimal requirements on recharging infrastructure hardware and site layout.

### 5.6.3. Assessment of EN 17161:2019 “Design for All - Accessibility following a Design for All approach in products, goods and services”

The Design for All approach means design for human diversity, inclusion and equality. Its aim is to enable maximum use of products, goods and services. This CEN-CENELEC standard is a consistent way for organizations to manage accessibility through a Design for All approach, so that products and services can be accessed, understood and used by all people.

The core mind-set and value of the Design for All approach in the EN17161:2019 standard is to encourage and help organisations to take human diversity into account in their design and development activities. The Design for All approach focuses on user involvement and experiences during the design and development process to achieve accessibility and usability. It should be applied from the earliest possible time, and throughout all stages in the life of products and services which are intended for mainstream use. A Design for All approach also focuses on user requirements and interoperability between products and services across the end-to-end chain of use to reach inclusive and non-stigmatizing solutions.

There are elements in the standard EN 17161:2019 that are relevant for recharging infrastructure for persons with reduced mobility, but the main aim of EN 17161:2019 is to give the market design principles, rather than a specification on recharging infrastructure. It covers the design and placement of recharging stations, including location spacing, surrounding environment, accessibility, and user information/signage without leading directly to minimal specified requirements.

The challenges of the use of the standard EN 17161:2019 regarding the accessibility of recharging infrastructure are:

- The standard is intended as a design guideline: this implies that extensive work should be done to make this design-for-all principle applicable as an EU wide standard for accessibility.
- The design-for-all principle offers flexibility in custom design processes for recharging station manufacturers and recharging station operators. EN 17161 is rather a framework for inclusivity than a technical design specification document. EN 17161:2019 includes recharging station dimensions, which might not be relevant for users with reduced mobility.

**In conclusion**, it is suggested that the necessary standard may include among others:

- a) technical specifications for both electric recharging and hydrogen refuelling (considering its concrete characteristics);

- b) hardware & pole/recharging station (e.g. ensuring that the buttons or screen of the recharging station are at an appropriate height and the weight (interval) of the recharging and refuelling cables so these are accessible);
- c) associated parking spaces and surrounding environment (e.g. providing sufficient space (range) for the parking place around a recharging/refuelling point or ensuring that the recharging point is not installed on a kerbed surface);
- d) data/information and software accessibility. In this respect, there is no overlapping with EN17161:2019, as it does not include harmonized technical design requirements, specific components (e.g. display/connector) or parking bay layouts.

It is also concluded that EN17161:2019 does not overlap with the development of a new standard for accessibility in alternative fuel infrastructure for people with disabilities. Specific clauses and Annexes can contribute and complement the new standard.

Summing up, all the existing standards and its analysis can serve as a basis for the development of a new standard for accessibility of recharging points even more that neither of them alone covers all the aspects of accessibility of recharging infrastructure.

## 6. Recommendations for minimal requirements

**Addressing the accessibility of recharging infrastructure for electric vehicles is a collective effort that should be done in close cooperation between public authorities, policymakers and industry stakeholders.** Even though the market share of vehicles for people with disabilities constitutes only a small percentage of the total vehicle market, it requires collective attention from public authorities and policymakers, infrastructure manufacturers, research institutions, engineering departments, healthcare organizations, and individuals with disabilities themselves. Collaborative efforts are crucial to develop better, more sustainable, and accessible solutions for people with disabilities.

AFIR is now in place and has established ambitious targets for EV recharging infrastructure deployment in the EU. In particular, AFIR Annex II, point 1.18 allows for the adoption of technical specifications for recharging stations to ensure access to users with disabilities. However, in the absence of concrete technical requirements for that infrastructure with regards to accessibility, there is a risk that the needs of persons with disabilities are not adequately accounted for rapidly deploying EV recharging infrastructure if the public authorities are not vigilant and proactive. Therefore, it is recommended that this transition is supported, guided, and potentially regulated by political institutions to ensure that the rights and needs of people with disabilities are duly considered and respected. “The ‘Enable Final report’ realised for UK Power Networks in 2022 contains several suggestions for a coordinated approach including knowledge sharing and a more proactive dialogue between public authorities and private stakeholders on plans and policies for EV recharging deployment.

We should aim to make recharging spaces accessible to all drivers, including persons with disabilities. However, this is a process that will take some time, and therefore these guidelines aim to provide support to public authorities and private stakeholders to ensure that at least an adequate number of locations are inclusive for all. Especially in the early stages of market development, the number of accessible charging spaces could be proportionate to the number of drivers with limited mobility and EV market share. These guidelines aim to provide support to public authorities and private stakeholders to ensure that at least an adequate number of locations are inclusive for all.

Although these recommendations apply mostly for newly-built recharging stations, whenever technical and financially feasible, they should also be considered when upgrading, converting or expanding existing recharging infrastructure for electric vehicles, while ensuring they are compliant with AFIR.

It is important to mention that, for the purpose of this document, only minimal requirements for site layouts of recharging points were specified. Further and more detailed specifications can be defined via the standardisation process. However, it should be noted that standards are always voluntary unless

mandated through legislation (e.g., AFIR). AFIR Annex II, point 1.18 allows for the adoption of technical specifications for recharging stations to ensure access to users with disabilities. At present, it is important to initiate work and support European Standardisation Organisations (ESOs) to develop relevant standardisation work and build consensus around it.

This chapter contains therefore recommendations for minimal requirements to be adopted in future EU standard on accessibility and other more general recommendations. The recommendations are based on existing practices from various guidelines and actual implementations in the field. The basis was the analysis in earlier chapters on compatibility and consistency between various national guidelines. It is hoped that until the delivery of the European Standardisation working support of AFIR Annex II, point 1.18, the presented recommendations can serve in the meantime as the inspiration and framework for public authorities.

As this report makes clear, an accessible recharging space consists of numerous different elements and the following recommendations delineate minimum requirements in each area however but is very challenging to establish a single or overarching definition of full ‘accessibility’ (e.g., a location might meet the minimum requirements for the height of control functions and cable, but the recharging point is on a kerbed instead of flat surface). Having this in mind, it should not prevent public authorities to pursue the objectives of ensuring that the EV recharging experience is as inclusive as possible for all.

Furthermore, the numerous elements which comprise an EV charging location are not all under the control or authority of the same entity. For example, the landowner may lease the CPO the parking space(s) and space for the charger(s), but not the space in between. So, if the chargers are on a kerbed or elevated surface, the CPO may not have the authority or permission to create a ramp, adjust bollards, or otherwise adjust the environment to enable accessibility. Likewise, the space in between could be separately governed by the municipality and require special permits, etc. These realities need to be understood in the issuance of recommendations and in the cost/benefit of making spaces accessible.

One fundamental element of achieving inclusive design in publicly accessible recharging stations is **user testing**, particularly with disabled individuals. The insights gained from initiatives where such testing was done, have underscored the paramount importance of the points of access, which refer to the areas immediately in front of charge point components. These access areas, along with factors such as reach, spacing, and visibility, play a pivotal role in ensuring the accessibility and usability of public recharging stations. To create a more accessible urban environment and promote the adoption of EVs, it is recommended that we address these challenges comprehensively. Beyond the design of the stations themselves, this can also be applied to the ease of use of mobile apps, which can be better personalised to fit each user’s needs. This is

highlighted in the German report *“Einfach Laden ohne Hindernisse: Anforderungen an barrierefreie Ladeinfrastruktur”*.

At the same time, upgrading and adaptation of EV charging infrastructure should be conducted based on such an approach that a balancing of different needs is adopted: on the one hand, the necessity to ensure access to EV charging infrastructure for all drivers, taking account different disabilities and cases of reduced mobility, and on the other, the necessity to ensure the practical feasibility and economic viability of upgrading projects of EV charging infrastructure.

In the following sections, these guidelines will provide recommendations on minimal common requirements that aim at improving the physical and digital interface of accessible recharging stations, technical parameters and standards, and various payment options, as well as the surrounding environment such as the parking space and associated amenities. The overarching aim is to enrich accessibility and usability for all users and to establish a framework, foundation and key elements for inclusion in a European Standard.

The recommendations are structured around the following topics:

- Hardware and parking space
- Data
- Distribution of Accessible recharging infrastructure
- Future Standards

## 6.1. Hardware and parking space

There is the need for an EU minimal level of requirements for recharging infrastructure, to avoid the proliferation of/national regional guidelines which lead to a fragmented ecosystem and regulations. The following topics should be considered as needing minimal requirements:

- All user interface elements or a recharging point (e.g. display, payment terminal, connector handle, grip holster) should be classified as operable parts, with a range of height dimensions that allow for its usage by all users, including persons on wheelchairs. Based on the currently published national guidelines, **the operable parts on a recharging point should be positioned within a range of 750 mm to 1300 mm height**. Special attention should be given to ensure that the minimum height of these components does not create unintended issues for non-disabled people, for example very tall persons.
- Whenever possible, parking spaces and the surrounding area should ensure free moving space for persons with reduced mobility. Based on the currently produced guidelines accessible recharging point requires

a **minimum parking space of between 2.200 and 3.500 mm wide and between 4.800 and 5.000 mm long**. In addition, the manoeuvring area as **free moving space shall be between 1.200 mm and 1.500 mm**.

- Where it may not be possible to apply the full dimensions (for example, on existing parking lots or kerbside parking and recharging), a 1.200 mm manoeuvring/access zone between the vehicle and the charger should be retained as a minimum.
  - However, there are many cases where providing this access zone over multiple parking space could reduce the overall amount of parking space in a given area. This could conflict with local parking policy or risk undermining the business case for EV recharging at this location. Balancing the need for accessibility with the number of charging spaces should be a key consideration.
  - This can be especially challenging in certain areas where existing parking spaces are being converted into 'recharging areas' - such as streetside parking in urban centres or parking garages, where space is tightly controlled. In these situations, there are stringent limits on applying a universal design principle, but this should not prevent the electrification of these locations in a way accessible for people with disabilities if public authorities make such requirements in public tendering.
- The recharging station's plug handle should ensure a secure grip without slipping. This includes considerations like surface finish and handle design.
- The recharging cable should be as light as possible while respecting relevant safety and operational requirements.
- In case of DC chargers where cables are attached to the station, the cable length should be sufficient to reach any vehicle, accommodating various socket positions however it should ensure that it doesn't obstruct the surrounding area during recharging, which could pose accessibility challenges for disabled individuals and others. A cable management system, such as overhead cable support, can be implemented to prevent the cable from obstructing kerbside or roadways and to prevent it from dragging across the ground.
- The sockets at recharging stations should be easily visible and visually distinctive for users regardless of their location.
- Regarding payment methods, the integration of contactless payment pads at recharging stations, provide a convenient and efficient method for everyone, in particular for disabled users to make payments. The AFIR requirements will contribute to this aspect.

- Public recharging stations should be installed on a flat, stable, and non-sloping surface with adequate grip to ensure safety and ease of use. A gradient not steeper than 1:50 could be recommended if necessary. It is preferable to position recharging stations on the same level as the roadway to facilitate easy access. However, in cases where height differences are unavoidable, disabled access features such as kerb drops and ramps, built in compliance with existing regulations, should be provided.
- This recommendation does not prevent the operator and/or owner of the recharging point to have ‘freedom of site design’ to optimize user experience for all users, including the design of individual elements (e.g. sidewalks, bollards, wheel stops, kerbs).
- Public authorities (when the recharging point is located in public space) and private entities (e.g. landowner, operator and/or owner of the recharging point when it is located in private area) are encouraged to guarantee barrier free access and to accommodate site layout adjustments to facilitate persons with reduced mobility on an equal level.
- Whenever possible, when installing any recharging pole/station, site owners should consider installing this equipment in close proximity to, for example, pedestrian crossings which are already accessible, or in already accessible/adapted kerbside. These simple solutions do not always require big investments, and can have a huge impact on users.
- As mentioned previously in this document, the accessibility requirements set out in Directive (EU) 2019/882 (European Accessibility Act) could be taken into account, where applicable, also for recharging and refuelling infrastructure.
- The operator and/or owner of the recharging point should ensure, where possible, that every Electric Vehicle type and every individual user have the ability to charge with sufficient cable reach. To this aim, the Operator should endeavour to position the recharging points in a way that every single vehicle can be reached without any obstacle for the user. This is particularly important for drivers with disabilities and elderly people driving EVs.
- Public authorities should consider whether the recharging stations can be placed next to or between the parking spaces instead of on the sidewalk, so that they do not create an additional obstacle or take up space for pedestrians.

## 6.2. Data requirements

In line with AFIR Art.20, and the recent Delegated and Implementing Acts, in particular Commission Implementing Regulation (EU) 2025/655, recharging point operators shall make available to the national access point (NAPs) the data types specified in that article at no cost.

Fulfilling this obligation with high-quality data will already significantly improve the accessibility of information for people with disabilities regarding the suitability of infrastructure.

In addition to this, the following recommendations should be considered in view of future work to enhance data availability regarding infrastructure suitable for people with disabilities:

- All data elements serving a better user experience should be channelled and adopted in a relevant and commonly accepted data exchange standard, in line with the format requirements set out in Article 1 of Commission Implementing Regulation (EU) 2025/655
- Future additional data types may cater the following data elements to improve user experience: 1) core accessibility data, 2) additional accessibility data, 3) general data points.
- CPOs should be able to report more information, and future European standards should help CPOs to better make available data to inform whether a recharging point is accessible or not for people with disabilities.

Additional accessibility data may optionally include:

- Parking space adapted for wheelchairs [Y/N]
- Further indication on the recharging station position [drop down] (e.g., Left, Front, Right)

The role of a future European standard will be key on the implementation of data on accessibility.. It should be the responsibility of operators/owners to prove this compliance y.

- Additional data should be captured in simple data elements. Too many details will diffuse end-customer information and will be a burden for quality checks and maintenance.

### 6.3. Distribution of Accessible recharging infrastructure

- Criteria like future demand based on actual usage rate of the recharging locations could be used to select the most suitable places for priority parking for disabled people, including installing recharging stations. In fact, under AFIR, publicly accessible recharging points cannot be reserved for people with disabilities.
- When a minimum number or percentage of accessible recharging points is specified, the definition of how this minimum number or percentage of accessible recharging points is distributed within the territory (e.g. region, city, aggregated urban area) should be done in a way that ensures equitable distribution across the territory (based on the planning of public authorities and included as elements in public tenders). For urban areas, this will in some cases require an analysis and distribution considered at street level.
- Public authorities have an important role in determining the distribution of the accessible recharging points at local level as they have knowledge of the current EV recharging demand (e.g. the presence of supermarkets, hospitals/medical centres/ other urban locations, fast recharging hubs) and on the potential demand for accessible recharging points based on the size of the population with disabilities (e.g. based on number of handicapped parking permits provided).
- The information about this distribution should be publicly available to users, CPOs and all other relevant stakeholders.
- This will ensure local authorities, as well as the private sector, have disabled consumers and accessibility considerations as a focus of their infrastructure strategy planning.
- For recharging locations with multiple recharging points, if a minimal required number of recharging points is specified (it means they are accessible for disabled people but not reserved), this can be done either based on a fixed number or percentage: minimal one (1) number of accessible recharging points per location with more than a specific number of recharging points or defined percentage of all recharging points per location. Based on the French experience, this percentage could be subjected to degressive rates depending on the number of parking and recharging places.

## 6.4. Future Standards

- Some STF members considered the utility of creating a standard on accessibility of recharging infrastructure. If implemented by stakeholders, it would facilitate for persons with disabilities the certainty about what they may expect at charging stations and it would bring benefits to manufacturers of recharging points by economies of scale.
- AFIR, Annex II, point 1.18 establishes the possibility to adopt technical specifications for recharging stations to ensure access to users with disabilities. In that respect the European Commission has issued a standardisation request to the CEN-CENELEC. The standardisation work led by CEN-CENELEC is expected to start during 2025 and should be aligned with existing practices in the EV recharging market.
- When developing these standards, standardization organisations together with the user groups supporting disabled persons, should address technical specifications for parking locations and recharging stations building on existing standards as mentioned above. That way communication protocols can refer to these standards. The standards should be based on use cases and overall requirements that keep freedom to individual companies to comply to it.
- It should also be noted that any accessibility standards should align with existing obligations placed on EV recharging companies including, but not limited to, the Alternative Fuels Infrastructure Regulation (AFIR), the Payment Services Directive (PSD2/3), the Energy Performance of Buildings Directive (EPBD) and the European Accessibility Act.
- When looking at the cost of retrofitting the existing EV recharging infrastructure, and the potential adjustments required to make them inclusive for all, in some cases it will not be economically feasible to have to change or retrofit existing recharging units based on accessibility standards. Additionally, in several European cities, in particular the ones with historical city centres, there are severe limitations in terms of availability of space to meet accessibility requirements of the EV recharging station surrounding areas. Standardisation initiatives should consider this.
- It is advised to detail accessibility requirements on the highest possible level and announce in a clear way by the public authorities so that the defined rules include all persons with disabilities and elderly people..

## 6.5. Other recommendations

- A clearer governance model is needed for a successful implementation of accessible recharging infrastructure, as this is shared responsibility between public authorities and private stakeholders. As mentioned previously, the elements which comprise an EV charging location are not all under the control or authority of the same entity, so a cooperation framework is needed. The STF should continue to play a key role in bringing all stakeholders around the table on this topic.
- Considering that the preparation of a new standard at the EU level can take 2-3 years, these guidelines could act as an intermediate step, so that public authorities have in the meantime certain examples of harmonised references that they can use in their policy.
- Disabled users of electric vehicles should be supported with respect to information services and providing feedback. This means ensuring that disabled users are properly aware of their recharging infrastructure options with respect to public and private recharging and have easy access to information on possible recharging solutions tailored to individual needs.
- Crowdsourcing tools could be used to gather insights on the user experiences of disabled drivers, particularly the issues they experience with electric vehicle recharging infrastructure, whilst also providing collective information on accessibility of public recharging points.
- In view of future requirements and obligations in terms of energy efficiency of charging infrastructure, the impact of accessibility elements such as bigger screens or brighter interfaces could be considered in the assessment of the energy consumption of EV recharging infrastructure.

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## ANNEX - Overview of authors, coordinators and review team

| Role                        | Organisation                 | Persons involved                                                    | Description of the role and organisation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Author                      | POLIS Network                | Pedro Gomes<br>Juliette Thijs<br>Zsofia Jakoi                       | POLIS ( <a href="http://www.polisnetwork.eu">www.polisnetwork.eu</a> ) represents more than 120 cities, regions and transport operators from all over Europe. POLIS' objective is to support European cities and regions to improve the quality of life of their citizens through innovative measures for sustainable urban transport. The Network facilitates access to European initiatives and research programmes for its members, looking into solutions for urban and regional mobility, in the field of health and environment, traffic management and intelligent transport systems, road safety, and social and economic aspects of transport. |
| Coordinator                 | DG MOVE, European Commission | Aleksandra Klenke<br>Dario Dubolino<br>Kai Tullius<br>Saki Gerassis |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Contributor & Core Reviewer | Vrije Universiteit Brussel   | Gabriela Barrera                                                    | VUB is an Urban Engaged University in Brussels, the heart of Europe. 20,000 students, nearly a quarter of them from abroad, are engaged in building their future and that of the world. With top-quality scientific research and customised education, VUB makes an active and engaged contribution to a better and more sustainable society.                                                                                                                                                                                                                                                                                                           |
| Contributor & Core Reviewer | EVRoaming Foundation         | Michel Bayings                                                      | The objective of the EVRoaming Foundation is to facilitate roaming services for charging electric vehicles and provide transparent information to consumers about charging locations and prices, by use of the open and independent Open Charge Point Interface (OCPI) protocol. The ultimate goal is to allow any EV driver to charge at any charging station in the EU: simplify, standardize and harmonize. The EVRoaming Foundation manages and maintains the OCPI protocol and ensure its free availability. The foundation is not limited to OCPI and can also support other related activities and services.                                     |
| Contributor & Core Reviewer | Tesla                        | Mattheo van der Molen                                               | Tesla is an American multinational company that specializes in electric vehicles, energy storage, and solar energy solutions. Founded in 2003, Tesla is known for its range of electric cars. Beyond vehicles, Tesla develops and sells energy storage products like the Powerwall and Megapack, as well as solar panels and solar roof tiles.                                                                                                                                                                                                                                                                                                          |

| Role                        | Organisation              | Persons involved                                 | Description of the role and organisation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|---------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Contributor & Core Reviewer | ChargeUp Europe           | Erno Scheers<br>Chiara Corradi<br>Aaron Fishbone | ChargeUp Europe is the voice of the EV charging industry, working towards the rapid deployment of charging infrastructure across Europe and a seamless charging experience for all EV drivers. Their mission is to push for an infrastructure rollout that starts from the consumer needs, make EV driving convenient across borders and to strive for an open-market model supported by open standards and protocols, and remove market barriers, for instance related to concessions, grids, data sharing, building codes, and more. |
| Contributor & Core Reviewer | European Disability Forum | Daniel Casas                                     | EDF is an umbrella organisation of persons with disabilities that defends the interests of over 100 million persons with disabilities in Europe. EDF is an independent non-governmental organisation (NGO) that brings together representative organisations of persons with disabilities from across Europe. We are run by persons with disabilities and their families. We are a strong united voice of persons with disabilities in Europe.                                                                                         |

